Sultanate of Oman

Nama Water Services Company SAOC

Consultancy Services for Design and Supervision for STP, Sewer and TE Networks Systems in Ibri

Tender No. T/2719110/2025

Concept Design Report

Revision 2

Renardet S.A. & Partners Consulting Engineers

Project 2621 · September 2026

Contents

Abbreviations and definitions

Term	Meaning
AAF	Average annual flow
APSR	Authority for Public Services Regulation
BOD	Biochemical oxygen demand
CAPEX / OPEX	Capital expenditure / operating expenditure
CESMM3	Civil Engineering Standard Method of Measurement, third edition
COD	Chemical oxygen demand
EIA	Environmental impact assessment
GIS	Geographic information system
LPCD	Litres per capita per day
MoHUP	Ministry of Housing and Urban Planning
NCSI	National Centre for Statistics and Information
NOC	No objection certificate
PAEW	Public Authority for Electricity and Water
PE	Population equivalent
Qadf	Average daily flow
Qpdf	Peak daily flow
SRT	Solids retention time
STP	Sewage treatment plant
TKN	Total Kjeldahl nitrogen
TSS	Total suspended solids
UTM 40N	Universal Transverse Mercator zone 40 North, WGS 84 datum

A note on the term TE

The Terms of Reference and this report use TE to mean treated effluent, that is the treated product of the sewage treatment plant. The NAMA design guidelines define TE as trade effluent and use TSE for treated sewage effluent. To avoid ambiguity, this report uses the term treated effluent in full, and TSE where a guideline value is quoted.[1]

[1] PAM-GUD-201, Table 1, page 16; PAM-GUD-203, page 13. Confirmation of the preferred convention is requested.

Executive summary

This report presents the concept design for the wastewater and treated effluent systems serving Ibri, together with the sewage treatment plant that will receive the collected flow. It records the data collected to date, the checks applied to that data, the design basis adopted, and the state of the work at the date of issue.

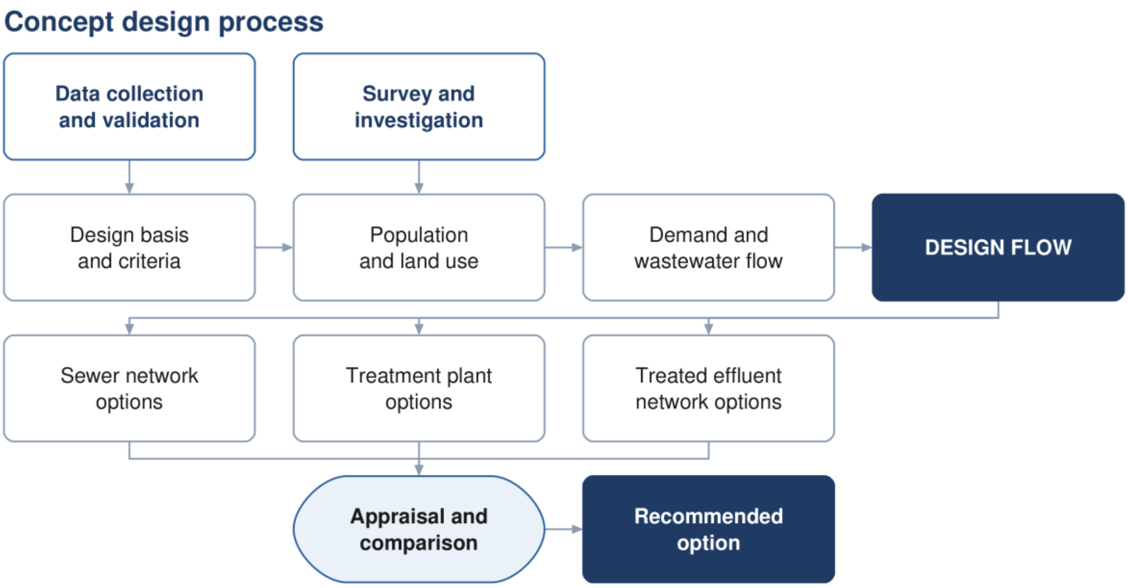

Figure 1 The concept design process, from the data collected to the recommended option.

The project area

The study area covers 531.4 square kilometres within the Wilayat of Ibri, Governorate of Adh Dhahirah, and contains twenty-five named settlements. All spatial work is carried out in UTM zone 40 North on the WGS 84 datum.[1]

Data collected

Data has been obtained from Nama Water Services covering the existing wastewater assets, the potable water network, electricity accounts and the cadastral plot layer. The datasets have been loaded into the project geographic information system, checked against the project boundary, and assessed for completeness. Section 7 records the outcome of those checks in full.

Four points arise from that assessment and are carried through the report.

- **Existing assets —** the wastewater dataset holds two networks. The constructed network, dated 2006, comprises 111.6 kilometres of gravity sewer and 10.0 kilometres of force main. A further 199.3 kilometres of gravity sewer, 23.2 kilometres of pumping main and the whole of the 45.7 kilometre treated effluent main are recorded as proposed. No treated effluent asset has been built.
- **Potable water —** the PAEW dataset provides 647.8 kilometres of water mains within the study area, and is adopted as the source for utility interfaces.
- **Electricity accounts —** 33,971 meters have been placed on the plots to establish the number of properties on each and its use.
- **Survey —** a topographic and utility survey covering the whole study area is in progress and will confirm levels, diameters and asset condition.

[1] Area measured from the project boundary supplied with the Inception Report. Two boundary versions were issued with that report; confirmation of the approved boundary is requested, and the item is listed in Section 5.

Design basis

Wastewater generation is derived from water demand in accordance with the NAMA design guidelines. For the Governorate of Adh Dhahirah the domestic consumption rate is 164 litres per capita per day, with uplifts of 22 per cent for non-domestic and 14 per cent for governmental consumption. Return rates of 85 per cent for domestic and tanker supply and 54 per cent for non-domestic supply are applied to obtain the wastewater flow.[1]

The occupancy rate is set for each settlement as its 2024 population divided by the domestic electricity meters counted in it, with a floor of four persons per property where institutional housing distorts the count and a cap at the highest rate among the larger settlements, 6.12; a settlement of fewer than a thousand people takes four. Ibri returns 6.07. Revision 1 used a single rate of 5.32 for the whole area; the rate per settlement replaces it. Section 14 sets out the derivation.[2]

Population and flows

Every plot in the study area now carries a population and an average sewage flow, for 2024 and for every year to the year its settlement is full. The use of each plot is read from the meters on it and from a satellite image of September 2026; the capacity of the empty land is read from the built plots around it; each settlement grows at the rate of the official series and fills its own land before overflowing to its neighbours. The study area holds 119,893 people in 2024 and generates 20,852 cubic metres of sewage a day. Ibri's land is full in 2056; the last settlement fills in 2070, with 349,029 people and 60,099 cubic metres a day. That is the saturation the Terms of Reference ask for, and the network is sized on it.[3]

Year	People	Average sewage flow, m³/d
2024, base	119,893	20,852
2030	139,535	24,216
2055	244,914	42,266
2070, saturation	349,029	60,099

Two industrial estates inside the town, at Al Tayyeb and Tanam, were found under the commercial tariff and are treated as special consumption. The army camp, a planned resort and two sources of tankered sewage outside the boundary are recorded in Section 16.[4]

State of the work

The design basis, the data assessment, the population and flow series and the assessment framework are complete. The network design has been developed and tested over a representative area of the town and is being extended to the whole study area as the survey data becomes available. The options for the sewer network, the treated effluent network and the treatment plant are presented as a framework in Part F and will be completed in the next revision.

[1] PAM-GUD-201, Table 11, page 60 and Table 19, page 71. The guideline states that the consumption values apply in the absence of updated figures and should be validated by NAMA before design.

[2] The guideline derives occupancy from population and housing units published by NCSI. Housing units are not published at settlement level, and counted domestic meters have been used in their place. The departure is recorded in Section 10.

[3] Average daily flows before infiltration and before the plant margin. The five-year series for every settlement is in Sections 14.7 and 15.7.

[4] The workforce of the two estates is an assumption, to be replaced by the estates' records.

How the options are developed and compared

Three options are developed for each of the sewer network, the treated effluent network and the treatment plant. Each set follows the character the guidelines describe: one advancing sustainability, one representing international best practice, and one based on practice already established in Oman. Every option meets the same functional requirement and the same effluent standard, so that the difference between them lies in how the result is achieved. Section 21 sets out what distinguishes them in design terms.

The options are compared over a twenty-five year period against seven criteria: total lifetime cost; sustainability, comprising carbon, circular economy and nature-based solutions; social development and in-country value; adaptability and resilience; operability; constructability; and environmental impact. Costs are discounted at five per cent. Nama Water Services sets the weight given to each criterion.

Where two options fall within ten per cent of one another on total lifetime cost they are treated as equivalent in cost, and the more sustainable of the two is adopted. Sensitivity is tested by varying the weighting between criteria, the discount rate, and the input design criteria.[1]

Deliverables

The Terms of Reference set out forty numbered deliverables for the concept stage. This report issues the design basis, the assessment of the data, the design criteria, the population and flow series to saturation and the framework for the options and their appraisal. The options themselves, the cost estimate and the comparison follow in the next revision, once the design horizon is confirmed and the survey is complete. Section 3.1 lists each deliverable and its position.

Matters requiring confirmation

Seven matters require confirmation from Nama Water Services. They are set out in Section 5 with the relevant references, and are summarised here. Section 10.1 lists the departures from the guidelines and the items awaiting data, among them tanker supply, private water sources and the connection ratio, for which confirmation is also requested.

	Matter
1	The design horizon to be adopted
2	The approved project boundary
3	Georeferenced versions of the project figures
4	The hydraulic modelling software to be used
5	The timing of the environmental impact assessment
6	The convention for the term TE
7	The design manual reference for site selection

[1] PAM-GUD-201, Sections 12.6 to 12.9, pages 104 to 106.

PART A

Project, scope and process

1 Introduction and background

Nama Water Services is developing wastewater collection, treated effluent distribution and sewage treatment for the Wilayat of Ibri. Renardet S.A. & Partners has been appointed to provide consultancy services for the design and supervision of these works.

This report is the concept design deliverable. It establishes the design basis, records and assesses the data collected, and sets out the framework within which options for the sewer network, the treated effluent network and the treatment plant are developed.

1.1 Objectives

The Terms of Reference set three objectives for the work: to verify the Regional Master Plan, to establish the ultimate expected sewage flow in the Ibri catchment, and to develop the concept, preliminary and detailed designs together with the tender documentation.

1.2 Structure of this report

The report follows the structure approved for the concept stage. Part A records the project and its process. Part B presents the data and the assessment of it. Part C sets out the design basis. Part D establishes demand and flow. Part E assesses the existing system. Part F presents the options. Part G carries the technical and financial appraisal, and Part H the delivery arrangements.

2 Scope and boundaries

2.1 The study area

The study area covers 531.4 square kilometres and contains twenty-five named settlements. Ibri is the principal settlement; the remainder are distributed along the wadi system and the main road corridors.[1]

[1] Area measured in the project geographic information system from the boundary supplied with the Inception Report, computed in UTM zone 40 North.

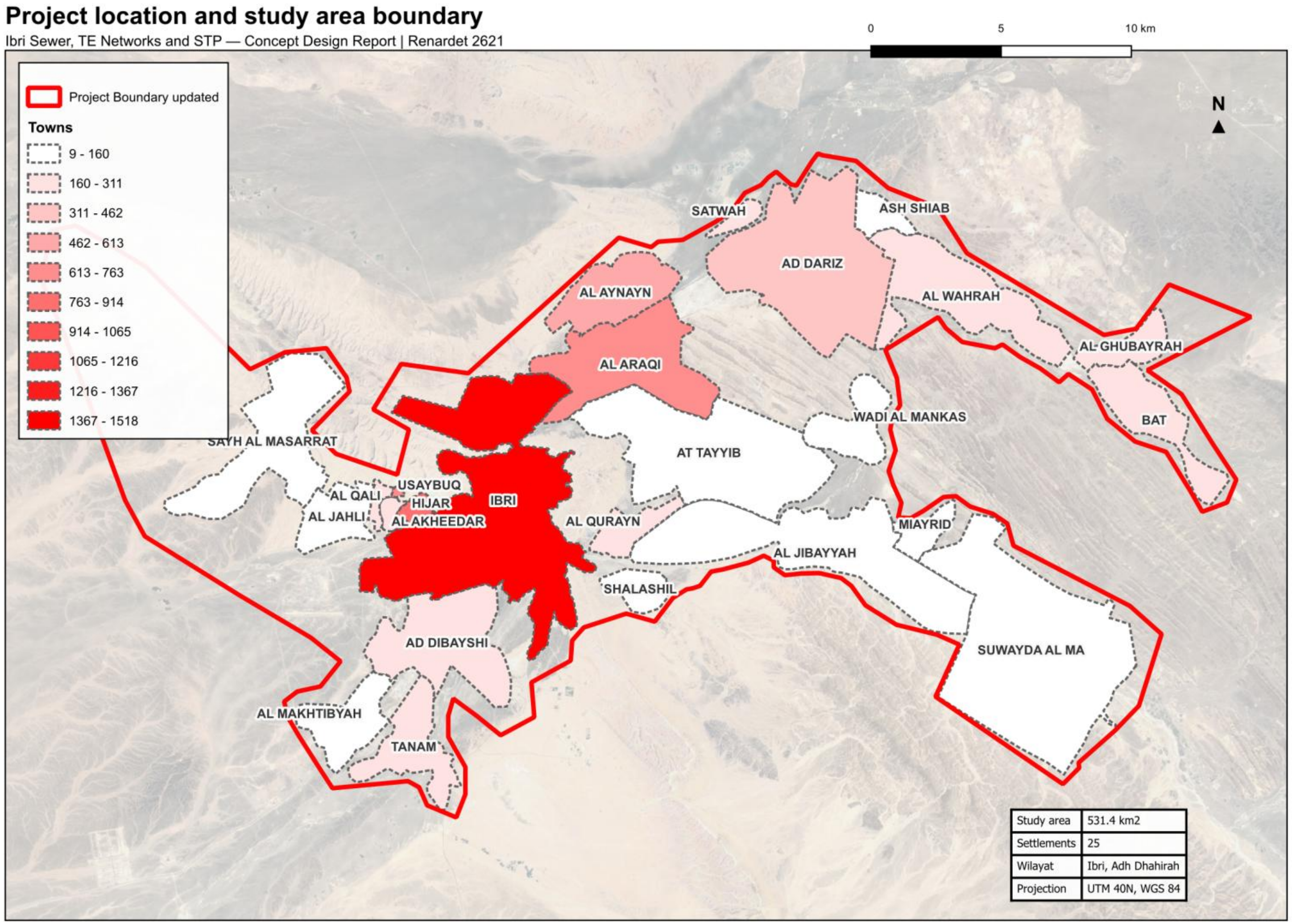

Figure 2   Project location and study area boundary, showing the twenty-five settlements within the Wilayat of Ibri.

2.2 Boundary

Two boundary datasets were issued with the Inception Report. The two differ in extent. The larger of the two has been adopted for the work presented here so that no part of the area is omitted, and confirmation of the approved boundary is requested.[1]

2.3 Coordinate system

All spatial data is held and all measurements are made in Universal Transverse Mercator zone 40 North on the WGS 84 datum. Datasets supplied in geographic coordinates have been reprojected. Levels are referenced to the terrain model described in Section 8.

3 Programme and design stages

The design comprises four stages: concept design, preliminary design, detailed design and the preparation of tender documents. Each stage is submitted for review and approval before the following stage begins.

The Terms of Reference allow sixty days for the concept design and a further twenty-one days for review. The preliminary design stage begins on approval of the concept design for the treatment plant.[2]

Two programmes have been issued during mobilisation, in the kick-off presentation and in the Inception Report. Confirmation of the governing programme is requested.

3.1 Deliverables at the concept stage

The Terms of Reference set out forty numbered deliverables for the concept stage. They are grouped below by subject. Those marked as issued form part of this report or accompany it; the remainder follow in the next revision as the survey and the options are completed.[3]

Table 1 Concept stage deliverables

Deliverable	Position
Executive summary and project schedule	Issued
Data collection report and assessment of the data	Issued
Design criteria and design basis	Issued
Population forecasting and flow projection at five-year intervals	Issued: population and flow per settlement at five-year intervals to saturation, and per plot
Topographic survey and geotechnical investigation	Survey in progress
As-built records and GIS for the existing systems	Follows the survey
Hydraulic assessment of the existing systems	Follows the survey
Wastewater network options, not fewer than three	Next revision
Treated effluent network options, not fewer than three	Next revision
Treatment plant options, not fewer than three, with siting and phasing	Next revision
Pumping and lifting station concept design	Next revision
Treated effluent and sludge management strategy	Framework issued; strategy in the next revision
Excess effluent and emergency overflow provisions	Next revision

[1] Project_Boundary.kmz and Final_Boundary_IBRI.kmz, both supplied in the Inception Report package.

[2] Appendix to the Form of Bid, page 147, restated as a contract term at page 177 of the tender document.

[3] Scope of Work, pages 63 and 64 of the tender document.

Deliverable	Position
Environmental impact assessment for the plant location	Follows confirmation of scope
Cost estimates and life cycle cost	Next revision
Risk analysis and value engineering	Next revision
Multi-criteria comparison and recommended option	Next revision
Hydraulic models in SewerGEMS and WaterGEMS	Follows confirmation of the software
Contracting strategy and implementation plan	Next revision
Register of approvals and no objection certificates	Maintained

3.2 Deliverables of the following stages

The preliminary design develops the approved concept to an estimate within ten per cent, with full survey, hazard and operability study, and preliminary bills of quantities. The detailed design completes the engineering, the drawings and the priced bills. Tender documentation follows the approved detailed design.

4 Consultation record

The following meetings have been held with Nama Water Services to the date of this report.

Table 2 Meetings held

Meeting	Date	Outcome
Kick-off meeting	15 July 2026	Project scope, approach, programme and organisation presented
Inception Report submission	August 2026	Design basis, methodology and programme submitted

Coordination with the authorities holding assets in the project area is described in Section 33, together with the register of approvals and no objection certificates.

5 Matters requiring confirmation

The following matters require confirmation from Nama Water Services. Each affects work that would otherwise be carried out twice, and confirmation is therefore requested at the earliest opportunity.

5.1 Design horizon

The Terms of Reference define the design horizon as the year of project completion plus twenty-five years, or the ultimate saturated flow. The Inception Report states that the planning and demand assessment extends to the year 2100. The two are different bases and produce different results. Confirmation of the horizon to be adopted is requested.[1]

5.2 Project boundary

Two boundary datasets were issued with the Inception Report, differing in extent. Confirmation of the approved boundary is requested, together with the extent to be surveyed.

5.3 Project figures

Figures 1, 2 and 3 of the tender define the project location, the areas requiring as-built records, and the areas subject to each design stage. They are supplied as images without coordinates. Georeferenced versions are requested so that the design areas can be established without ambiguity.

5.4 Hydraulic modelling software

The Scope of Work requires the wastewater network to be modelled in SewerGEMS and the treated effluent network in WaterGEMS. The staffing schedule in the tender refers to a different package. Confirmation of the software to be used for the deliverable models is requested.[2]

5.5 Environmental impact assessment

The design guidelines place environmental impact assessment scoping at the preliminary design stage in one clause and the full assessment at the concept and preliminary stages in another. Confirmation of the scope required at this stage is requested.[3]

5.6 Terminology

Confirmation is requested that the term TE, as used in the Terms of Reference and in this report, is understood to mean treated effluent.

5.7 Design manual reference

The Terms of Reference require the treatment plant location report to follow item 2.1 of Section 05 of the Wastewater Design Manual. The current revision of PAM-GUD-203 consolidated and renumbered the former manuals, and the corresponding content is now Section 10.1 of PAM-GUD-203, Site Selection. The report follows that section, and confirmation is requested.[4]

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[1] Terms of Reference, page 51 of the tender document; Inception Report Revision 0, Section 6.2.

[2] Scope of Work, pages 56, 57, 62, 65 and 73; Bidding Form 24, page 123 of the tender document.

[3] PAM-GUD-201, Table 2, pages 19 to 22, and Section 6.1.4.3, page 44.

[4] PAM-GUD-203 Revision 01, page 2, records the consolidation of the former manuals. Site selection is at pages 63 and 64.

PART B

Data

6 Data collection

Data has been requested from Nama Water Services and from the authorities holding assets in the project area. The table below records what has been requested and the position at the date of this report.

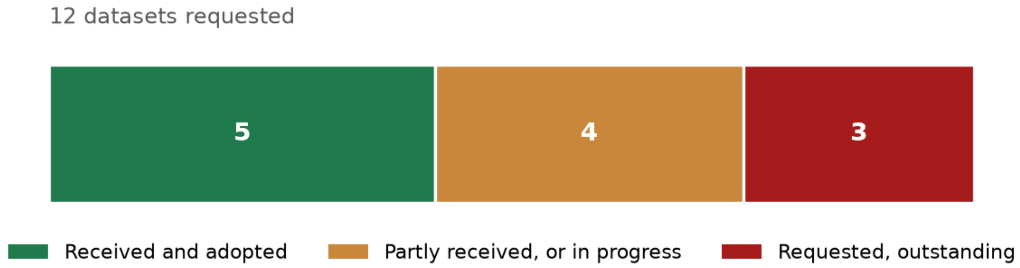


Figure 3 Position of the data register at the date of issue.

Table 3 Data register

Dataset	Received	Position
Constructed sewer network, as-built	Yes	Geometry received, dated 2006. Levels and diameters not recorded; being surveyed
Constructed force main	Yes	Received, dated 2006
Treated effluent network	Proposed only	The dataset contains no constructed treated effluent asset
Lifting station details	Partial	One station located; equipment and duty details being surveyed
Treatment plant records	Partial	Location and nominal capacity received; operating data outstanding
Electricity account data	Yes	Received
Cadastral plot data	Yes	Received. Land use and missing plots being addressed
Potable water network	Yes	Received as part of the PAEW dataset
Topographic survey	In progress	Survey team mobilised
Integrated Master Plan	No	Requested
Plant inflow and tanker records	No	Requested
Electricity, telecom and gas service records	No	To be requested from the respective owners

A topographic and utility survey covering the whole study area is in progress. It will establish cover and invert levels, diameters, materials and condition for the existing sewer, force main and treated effluent networks, together with the lifting stations, the topography and the cadastral boundaries including plot gates. The results will be incorporated in the next revision of this report.

7 Assessment of the data

Each dataset has been loaded into the project geographic information system, reprojected where necessary, and checked against the project boundary. This section records the quantity of each dataset, the proportion falling within the study area, and the limitations identified.

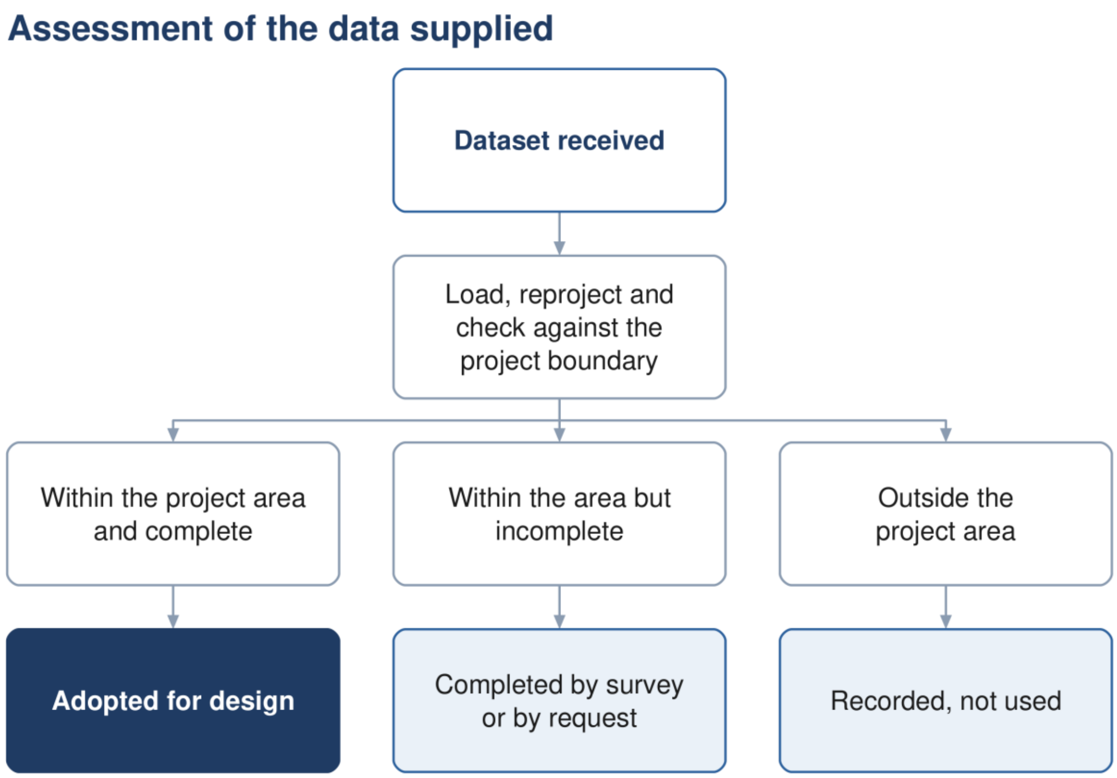


Figure 4 The assessment applied to each dataset supplied.

7.1 Wastewater assets

The wastewater dataset supplied by Nama Water Services holds two networks, not one. They are distinguished by the operational status field, and the distinction is confirmed by four further fields that agree with it on every record.[1]

Table 4 The two networks and how they are distinguished

	Constructed network	Proposed network
Operational status	1	0
Installation date	1 January 2006	not recorded
Source	drawings, and closed-circuit television	asset planning
Project code	5A-1 to 5A-5	SUREKHA
Remark	reference data	large urban area gravity, pumping main and treated effluent networks

On that basis the assets within the study area are as follows.

[1] Field OP_STATUE on SEWERLINE_IBRI, FORCEMAIN_IBRI and TE_LINE_IBRI. Value 1 denotes the constructed network and value 0 the proposed network.

Table 5 Wastewater assets within the study area, by status

Asset	Constructed features	Constructed length	Proposed features	Proposed length
Gravity sewer	3,267	**111.6 km**	129	199.3 km
Force main	1	**10.0 km**	8	23.2 km
Treated effluent main	none	**none**	8	45.7 km
Pumping station	1	**1**	—	—
Treatment plant	1	**1,800 m3/d**	1	29,038 m3/d, shown as design

The distinction is material to the design. The constructed network extends to 111.6 kilometres of gravity sewer and 10.0 kilometres of force main, serving the central part of the town, and discharges through a single lifting station to the existing treatment plant. It is twenty years old and its condition is not recorded. The remaining 199.3 kilometres of gravity sewer, 23.2 kilometres of pumping main and the whole of the 45.7 kilometre treated effluent main shown in the dataset are proposed, not built. No treated effluent asset has been constructed.[1]

Two limitations apply to the constructed network. Neither diameter nor invert level is recorded on any constructed gravity segment, so the hydraulic capacity of the existing system cannot be established from the data supplied. The client's own records carry a remark that the data is not reliable and is to be used for reference only. The survey now in progress will establish diameters, levels and condition, and until it reports, the capacity of the existing network to accept additional flow is treated as unknown.[2]

[1] Lengths measured within the approved project boundary in EPSG:32640. The proposed alignments are recorded in the dataset as an earlier planning proposal and are not adopted as design input; they are shown for reference and to identify any commitment already made by the client.

[2] Field REMARKS on the constructed records. The tender records that the existing network layout is based on available information, and that the preparation of complete as-built records and GIS forms part of the consultant's scope.

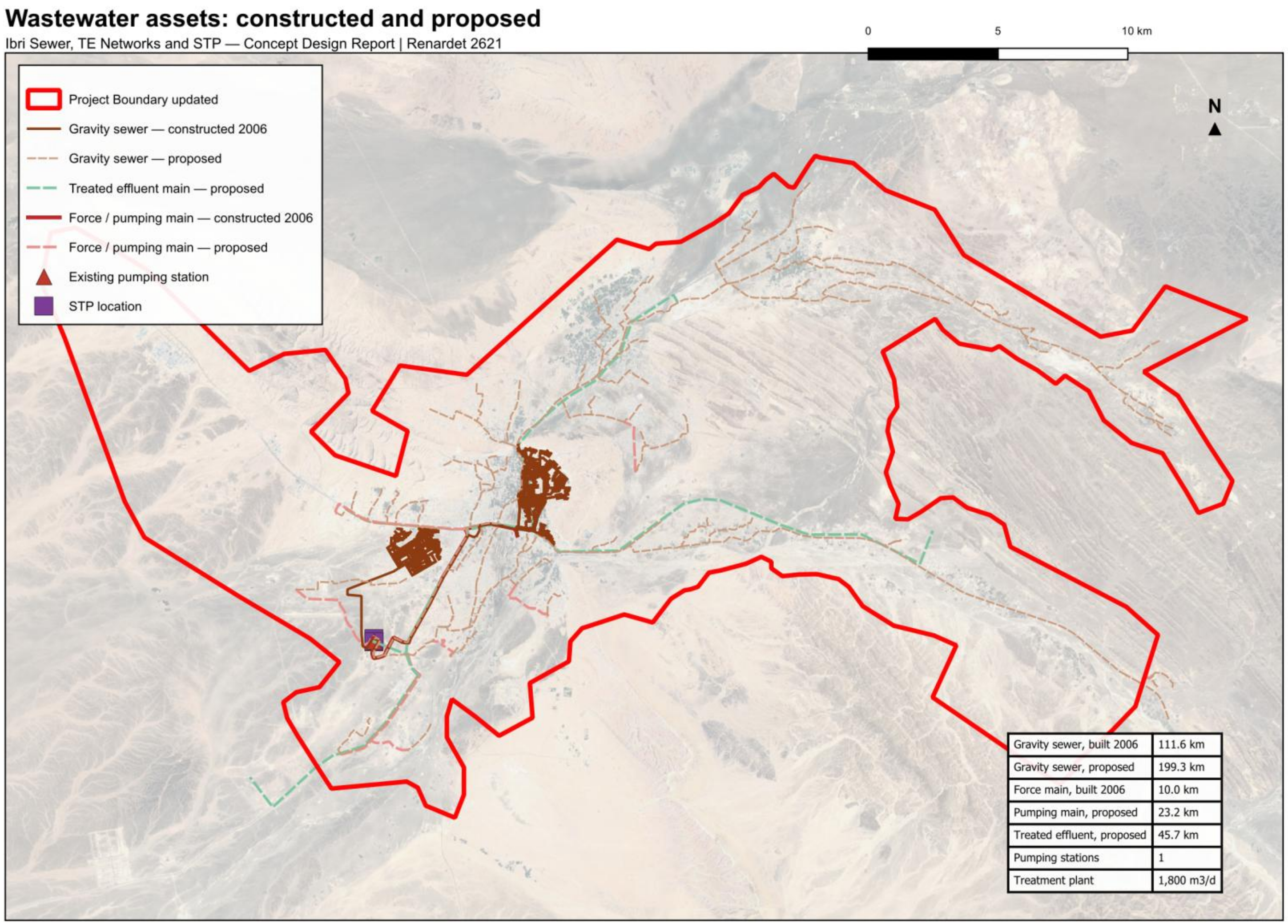


Figure 5 Wastewater assets within the study area. The constructed network, dated 2006, is shown distinctly from the alignments recorded in the dataset as proposed.

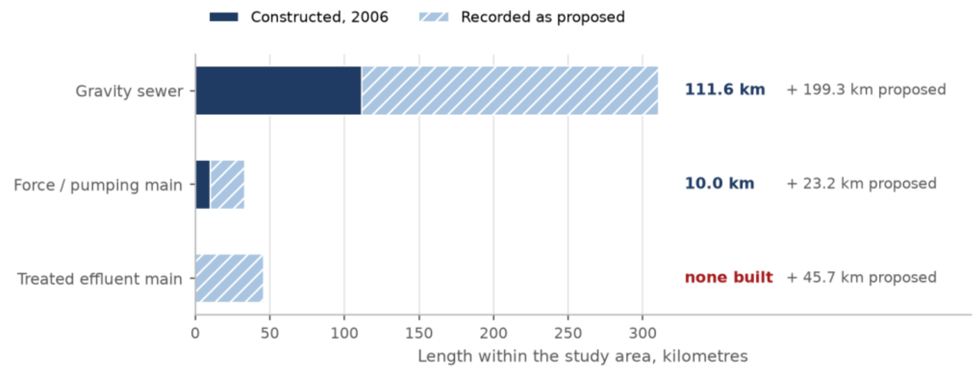


Figure 6 Length of wastewater asset within the study area, separated into constructed and proposed.

7.2 Potable water network

Two datasets describing potable water assets were supplied. They differ substantially in coverage.

Table 6 Potable water datasets

Dataset	Features	Quantity	Within the study area	Observation
PAEW water mains	6,156	1,315.9 km	647.8 km (49 %)	adopted as the source for utility interfaces
PAEW water laterals	1,954	12.7 km	8.1 km (64 %)	—
PAEW system valves	2,866	2,866 points	1,586	—
PAEW hydrants	965	965 points	568	—
PAEW services	318	318 points	—	—
PAEW facilities	265	265 points	20	—
NAMA water mains extract	90	3.5 km	none	located near 55.80 E, 24.27 N, approximately 130 km north-west of the project area

The PAEW dataset provides 647.8 kilometres of water mains within the study area and is adopted as the source for utility interfaces. The second dataset, supplied under an Ibri file name, contains 3.5 kilometres of mains located approximately 130 kilometres north-west of the project area; none of it falls within the study area. Confirmation is requested that the PAEW dataset is the current record for Ibri.[1]

[1] The extent of the second dataset is 55.802 to 55.814 degrees east and 24.269 to 24.292 degrees north, in the vicinity of Al Buraymi.

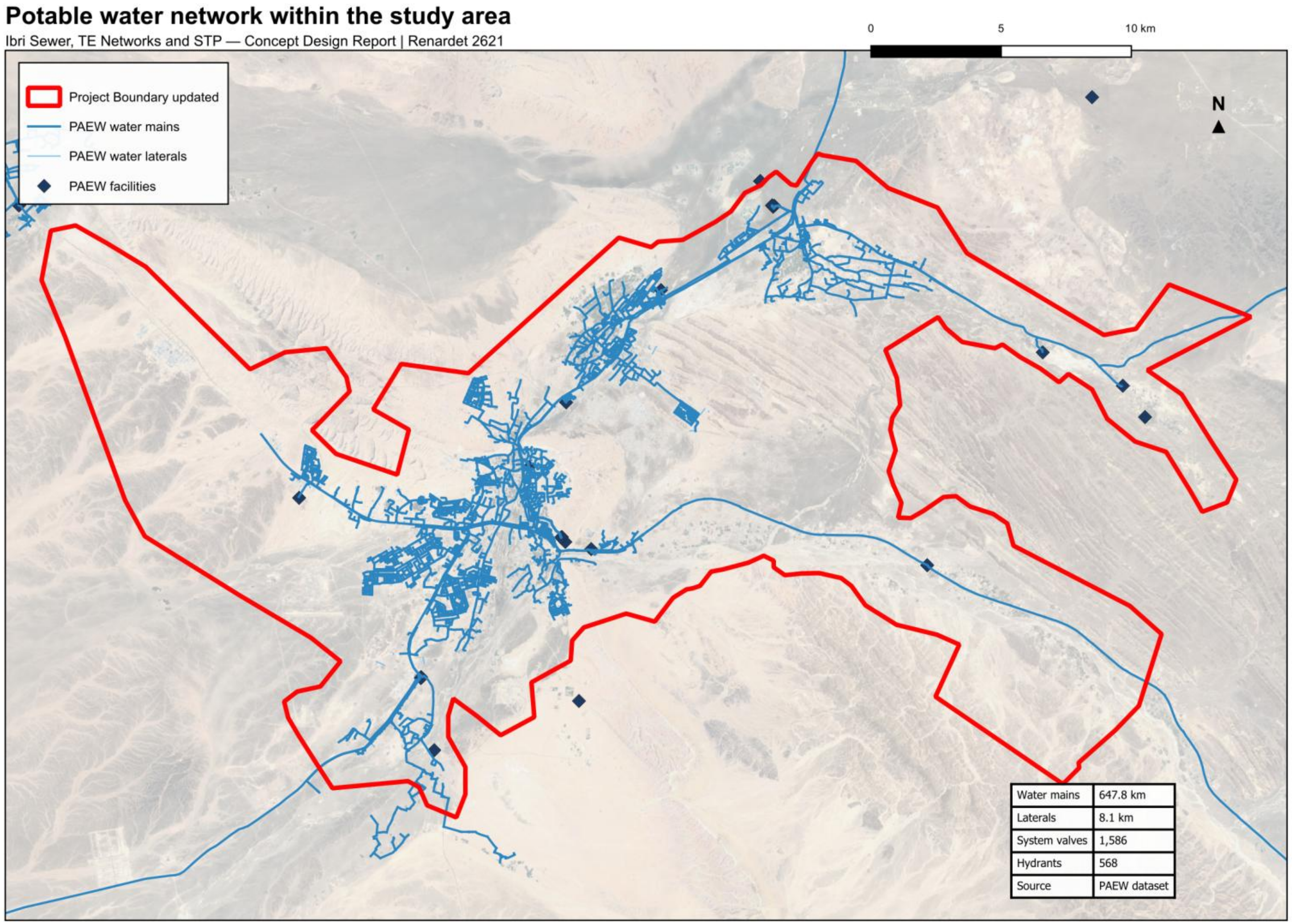


Figure 7 Potable water network within the study area, from the PAEW dataset.

7.3 Electricity accounts

The electricity account dataset contains 33,971 records, each carrying a tariff name and a coordinate. It records neither land use, floor area nor consumption, and the wilayat field is empty on every record. It establishes the number and category of connections at a location, and is used for that purpose in Section 14. The data is of 2024, which is the base year of this report.[1]

Table 7 Electricity accounts by tariff and the category adopted

Tariff	Accounts	Category adopted
Primary Account	10,973	Domestic: one property
Primary Account with National Subsidy	5,272	Domestic: one property
Additional Account	6,344	Domestic: one property, an additional dwelling
Commercial, Fisheries, Tourism	9,392	Non-domestic
Government, Defence	967	Governmental
Agricultural	523	Agricultural: an irrigation pump, no sewage
Industrial	1	Special consumption
Cost Reflective Tariff	499	Large consumer; use established in Section 16
Total	**33,971**	

The Cost Reflective Tariff is applied to consumers above a consumption threshold and is therefore a measure of size rather than of use. The 499 accounts carrying it have been placed one by one against public records, and each carries the use found and the evidence for it. Section 16.1 gives the result.[2]

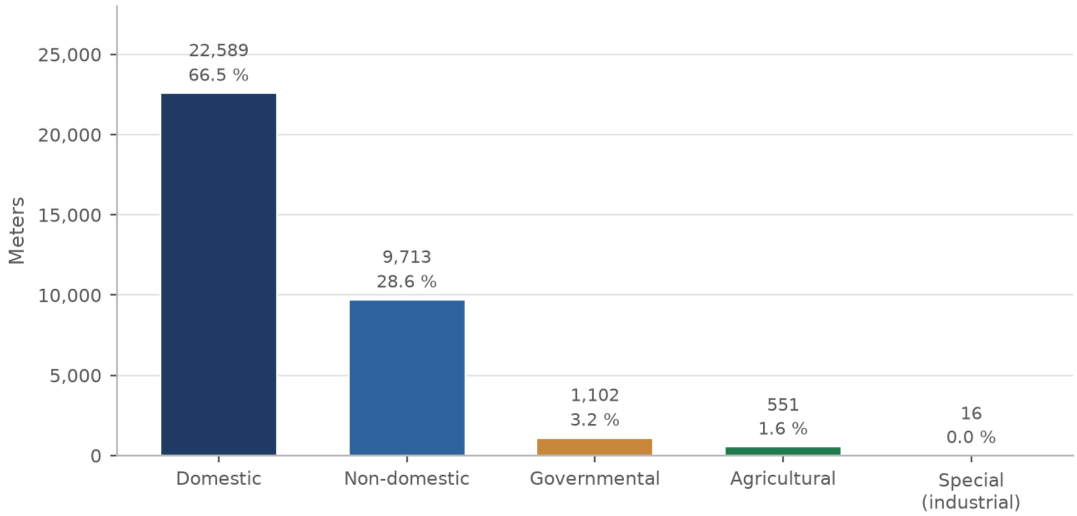


Figure 8 Electricity meters by the category adopted, after the large consumers were placed. Domestic connections, including additional dwellings on the same plot, are two thirds of the total.

[1] Fields present: identifier, tariff, coordinates in projected and geographic form, governorate and wilayat. The governorate is recorded as Dahira on all records; the wilayat field is empty on all records. The positions were adjusted onto the cadastre by the client before issue.

[2] The tariff comprises three variants in the dataset: fixed rate (9 accounts), seasonal (298) and time of use (192).

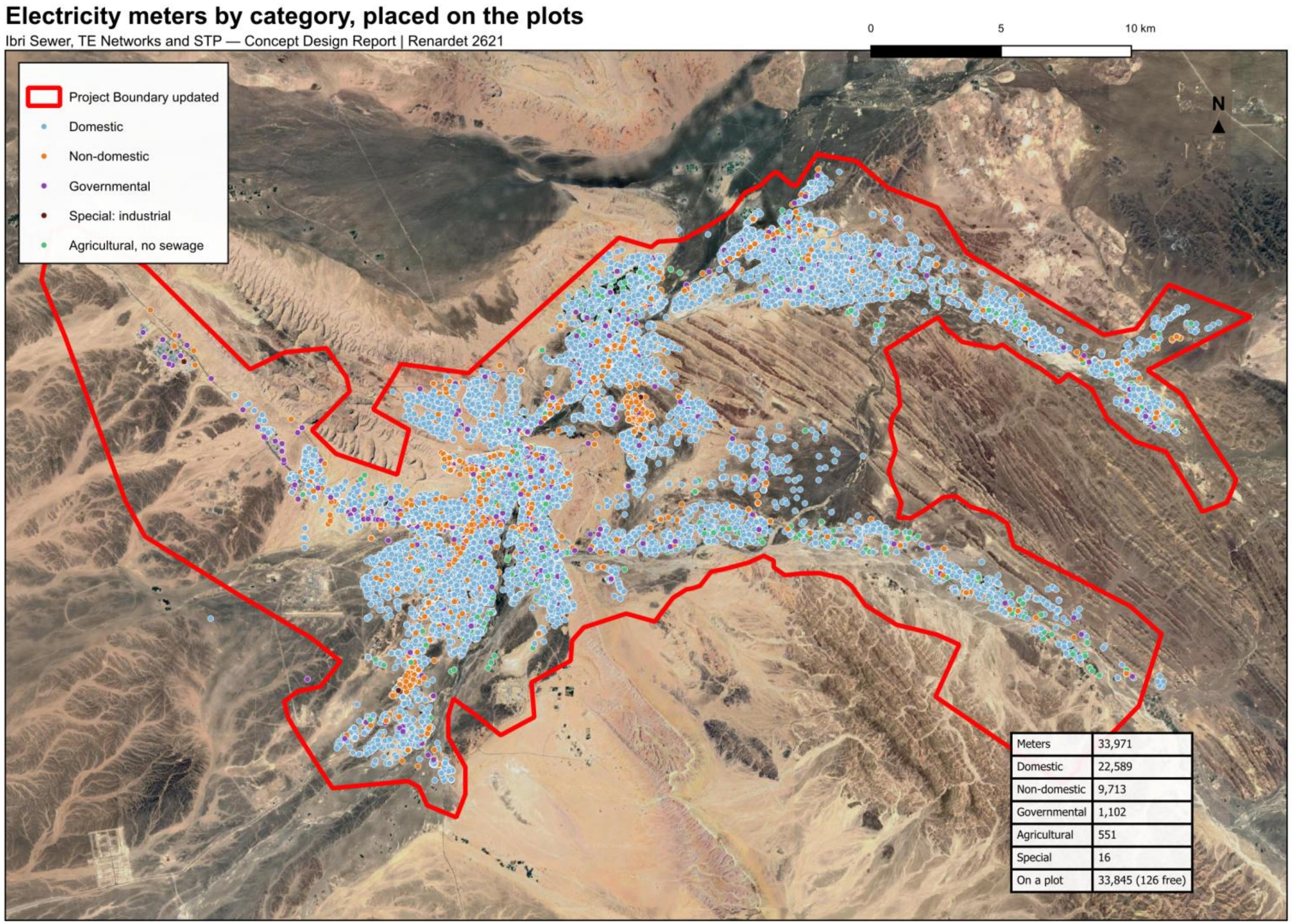


Figure 9   Electricity meters by category, placed on the plots. The pattern of connections defines the developed extent of each settlement.

7.4 Cadastral and settlement data

The cadastral plot layer supplied by the Ministry of Housing and Urban Planning, in the issue of September 2026, holds 77,265 plots with a built-or-future flag and a land-use class. The flag follows the electricity meters and is reliable. The class is not: it carries no government category, records whole districts under codes that mean unclassified, and disagrees with the meters on one built plot in seven. The use of each plot is therefore derived from the meters and the satellite, as Section 14.5 sets out. The survey now in progress will establish the cadastral boundaries including plot gates.

The settlement boundaries supplied with the Inception Report outline the built cores and leave the land between them unassigned. Section 14.2 describes how every plot has been assigned to a settlement and the boundaries redrawn to meet.

7.5 Other datasets supplied

Table 8 Further datasets received

Dataset	Extent	Observation
Electricity accounts	33,970 points	Tariff, coordinates and governorate. No land use, floor area or consumption is recorded.
Ibri Bypass scheme	13,404 placemarks	A drawing export carrying 11,715 sublayers of blocks, hatches and dimensions. Suitable as a positional reference; conversion is required before it can be used as structured data.
Al Raybah pipelines	3 files	Pipeline alignments at 110, 180 and 225 mm, within the project area.
Rehabilitation packages	6 archives	Work-order submissions covering Al Sad, Khadil, Yanqul, Dhank and Hay Al Aqabah, comprising drawings, meter schedules and geodatabases.
House connection packages	4 archives	Contractor submissions dated May and June 2026.

Five of the datasets supplied relate to areas outside the project boundary and have not been used. They are recorded here for completeness.

Table 9 Datasets relating to areas outside the project

Dataset	Extent	Location
Mudhaibi zone 6	544 placemarks	Wilayat of Mudhaibi, Ash Sharqiyah
Mudhaibi zone 8	461 placemarks	Wilayat of Mudhaibi, Ash Sharqiyah
Qabil Al Rukha	5 placemarks	Wilayat of Mudhaibi
Qunaib	4 placemarks	Wilayat of Mudhaibi
New location	8 placemarks	Approximately 30 km north of the project area

8 Survey and investigation

8.1 Topographic and utility survey

A survey team is mobilised and working across the study area. The survey covers the existing sewer network, lifting stations, force mains and treated effluent network as built; the topography; and the cadastral boundaries including plot gates. Its outputs will establish the levels, diameters and condition that the supplied datasets do not carry, and will be incorporated in the next revision of this report.

8.2 Terrain model

A bare-earth terrain model at 0.5 metre resolution covering the study area is in use for the work presented here. It will be superseded by the topographic survey for design purposes.[1]

8.3 Geotechnical investigation and trial pits

Geotechnical investigation and trial pits form part of the scope. Fifty trial pits are to be carried out at critical locations proposed by the consultant and approved by Nama Water Services. The programme of pits will be set out following the desk study of utility records described in Section 33.

9 Existing systems: as-built records and GIS

The preparation of as-built records and geographic information for the existing sewer and treated effluent systems forms part of the scope of work. The datasets supplied provide the geometry of those systems; the survey in progress will establish the levels, diameters, materials and condition required to complete the records.

The records will be prepared to the Nama Water Services specification and uploaded to the client's geographic information system for acceptance. Progress will be reported in the next revision.

[1] The model excludes buildings and is held in UTM zone 40 North.

PART C

Basis of design

10 Codes, standards and departures

The design follows the Nama Water Services design guidelines and standard specifications. Where a value is taken from a guideline it is cited to the guideline and page.

Table 10 Governing documents

Reference	Title
PAM-GUD-201	General Design Guidelines, Revision 01, March 2026
PAM-GUD-202	Water and TSE Design Guidelines, Revision 01, March 2026
PAM-GUD-203	Wastewater Design Guidelines, Revision 01, March 2026
MD 145/1993	Ministerial Decision, treated effluent and sludge reuse
MD 159/2005	Ministerial Decision, discharge to the marine environment
MD 41/2017	Ministerial Decision, ambient air quality

10.1 Departures

The departures from the guidelines that arise from the data available, and the items that await data, are set out below with the reason and the position adopted; confirmation is requested.

Table 11 Departures from the design guidelines

Subject	Guideline position	Position adopted	Reason
Occupancy rate	Population divided by housing units, both from NCSI	Settlement population divided by the domestic electricity meters counted in the settlement, with a floor and a cap	Housing units are not published at settlement level
Non-domestic and governmental demand	Unit rates per pupil, bed, employee and floor area where detailed land use allocation is available	The published governorate ratios	The quantities the unit rates require are not recorded in any dataset held
Allocation of non-domestic demand	Distributed across the served population	Placed on the plots whose meters generate it	A residential street generates no commercial flow; the total is unchanged and only its distribution differs
Industrial estates	Determined case by case with the developer	An assumed workforce at the dry-industry unit rate	No workforce or discharge record has been supplied; the assumption is stated and is to be replaced by the record
Water supplied by tanker	Assessed from the tanker filling station records	Households on tanker supply carry the domestic rate like any other metered household; no separate tanker term	No filling-station record is held; requested
Other water sources	Private wells and other non-network abstraction assessed	Not assessed; every flow is on the network-accounted basis	No record of private abstraction is held; requested

Subject	Guideline position	Position adopted	Reason
Connection to the sewer	Coverage rising to the whole served area by the end of the period	Every metered property counted as connected from 2024; pipes and the plant are sized on the full load, and the early-year self-cleansing case takes the connection ratio of the Inception Report	Sizing on the full load is the safe side; the ratio is carried separately where under-estimation would be the risk
Design flow standard	Where detailed land use is available, design flows calculated to a stated international standard such as BS EN 752	The planning ratios and return rates of the guideline, applied plot by plot	The quantities a detailed method requires are not held; the standard will be stated when they are
Growth beyond the forecast	Not extrapolated more than ten years beyond the available forecast	The Inception Report series to 2100, used for its growth rates only	The land fills after 2050; the saturation year rests on the capacity of the cadastre, and the horizon to 2100 was instructed by Nama Water Services

The unit rates will be adopted for non-domestic and governmental demand as soon as the quantities they require become available.[1]

11 Level of service and resilience

The systems are designed to convey and treat the flows arising across the design horizon without surcharge in the collection network and without loss of treatment capacity at the plant.

- **Redundancy —** sufficient capacity is provided that the design peak flow can be achieved with any one unit out of service.
- **Design margin —** a margin of ten per cent is applied to the treatment plant design flow, over and above redundancy.
- **Flood resilience —** structures are sited and set above the flood levels described in Section 29, and the plant is to remain operational during floods.
- **Failure —** emergency provisions are described in Section 26.

[1] PAM-GUD-201, Table 12, page 61. The rates are expressed per pupil, per bed, per employee and per square metre of floor area.

12 Design criteria: collection network

The criteria below govern the gravity network. All are taken from the wastewater design guideline.

Table 12 Gravity sewer design criteria

Criterion	Value	Reference
Self-cleansing velocity	not less than 0.75 m/s at peak flow	PAM-GUD-203 p26
Preferred velocity	0.90 m/s at peak flow	PAM-GUD-203 p26
Maximum velocity	3.0 m/s at the design depth of flow	PAM-GUD-203 p27
Depth of flow, up to 350 mm	0.65 of the diameter at peak flow	PAM-GUD-203 Table 10
Depth of flow, above 350 mm	0.50 of the diameter at peak flow	PAM-GUD-203 Table 10
Roughness, Colebrook-White	1.5 mm for all sizes and materials	PAM-GUD-203 p24
Minimum cover	1.3 m to the crown of the pipe	PAM-GUD-203 p33
Minimum cover with protection	0.5 m	PAM-GUD-203 p33
Recommended maximum cover	approximately 10 to 12 m	PAM-GUD-203 p33
Minimum diameter, laterals and mains	200 mm outside diameter	PAM-GUD-203 Table 6
Maximum lateral length	45 m	PAM-GUD-203 Table 6
Manhole spacing, 200 to 315 mm	100 m	PAM-GUD-203 Table 12
Manhole spacing, 350 to 900 mm	120 m	PAM-GUD-203 Table 12
Backdrop required	where inverts differ by more than 600 mm	PAM-GUD-203 p30
Inlet angle at a manhole	not less than 90 degrees to the flow	PAM-GUD-203 p19 and p30

12.1 Hydraulic formulation

Full-bore velocity is computed by the Colebrook-White equation.

$$V = -2\sqrt{2gDS}\log_{10}\left[\frac{k_s}{3.7D} + \frac{2.51\nu}{D\sqrt{2gDS}}\right] \quad (1)$$

Symbol	Meaning	Unit
V	full-bore velocity	m/s
g	acceleration due to gravity	m/s^2
D	internal diameter of the pipe	m
S	hydraulic gradient	m/m
k s	roughness coefficient, 1.5 mm	m
v	kinematic viscosity of the sewage	m^2/s

Gradients are set so that the self-cleansing velocity is achieved at peak flow, and are checked against the minimum gradients tabulated in the guideline. The tabulated gradients correspond to a full-bore velocity of 0.75 m/s; the velocity at the design depth of flow is verified separately for each run.[1]

12.2 Sediment transport

Two checks are applied together, and the steeper gradient resulting from them governs: the self-cleansing velocity above, and the minimum tractive force. At the head of the system, where the self-cleansing velocity cannot be achieved, the gradient is set by the tractive force method.

[1] PAM-GUD-203, Table 11, page 29. The tabulated values have been reproduced using the Colebrook-White equation with a roughness of 1.5 mm at 15 degrees Celsius, and correspond to full-bore flow.

$$S_{\min} = K\tau^{1.23}Q^{-0.461} \quad (2)$$

Symbol	Meaning	Unit
S min	minimum gradient to move the deposited particle	m/m
τ	tractive tension	Pa
Q	flow	m^3/s or l/s, with K taken to suit
K	coefficient, 2.33×10^{-4} for Q in m^3/s	—

The value of tractive tension to be adopted is not stated in the guidelines. A value will be proposed with its basis and confirmation requested before the gradients are fixed.[1]

[1] The equation and its coefficient are given at PAM-GUD-203, page 27. No corresponding value of tractive tension in pascals is stated in PAM-GUD-203 or PAM-GUD-201.

13 Design criteria: treatment plant

Table 13 Treatment plant design criteria

Criterion	Value	Reference
Design horizon	not less than 15 years	PAM-GUD-203 p65
Planning life cycle	25 years	PAM-GUD-201 p57
Design margin	10 per cent	PAM-GUD-201 p73
Size category	large where 20,000 m^3/d or above	PAM-GUD-203 p65
Organic load	not less than 60 g BOD per person per day; the measured strength of the existing works governs once received	PAM-GUD-203 p74
Solids load	not less than 80 g suspended solids per person per day	PAM-GUD-203 p74
COD to BOD ratio, domestic	1.8 to 2.2	PAM-GUD-203 p74
Effluent standard	Class A of MD 145/1993	PAM-GUD-203 p69
Total nitrogen	less than 15 mg/l as N	PAM-GUD-203 p71
Chlorine residual at the plant discharge	0.3 to 1.0 mg/l	PAM-GUD-203 p71
Chlorine residual at the consumer	more than 0.3 and less than 1.0 mg/l	PAM-GUD-203 p130
Treated effluent produced	95 per cent of the inflow	PAM-GUD-201 p73
Sludge produced	0.25 kg per cubic metre of inflow, indicative	PAM-GUD-201 p78

The total nitrogen limit is the governing criterion for process selection. Class A of Ministerial Decision 145/1993 does not itself set a total nitrogen limit, and the individual nitrogen limits it does set would together permit a higher concentration than 15 mg/l. Full nitrification and denitrification are therefore required.[1]

[1] Class A permits ammoniacal nitrogen at 5 mg/l, organic nitrogen at 5 mg/l and nitrate at 50 mg/l as NO_3, equivalent to 11.3 mg/l as N.

PART D

Demand and flows

14 Population and land use

14.1 Approach

The population of the study area, its distribution and its growth are established by counting rather than by assumption. Every domestic electricity meter is a property. Every plot takes its use from the meters on it. Every empty plot takes its future from the built plots around it. The official population series supplies the growth rate of each settlement and nothing else. The result is a population and an average sewage flow for every plot, today and in every year to the year the land is full, which is the saturation the Terms of Reference ask for.

The distribution of population across the study area is shown below from the Global Human Settlement population grid. The grid is a third-party product and is used to show where population lies, not how much of it there is.[1]

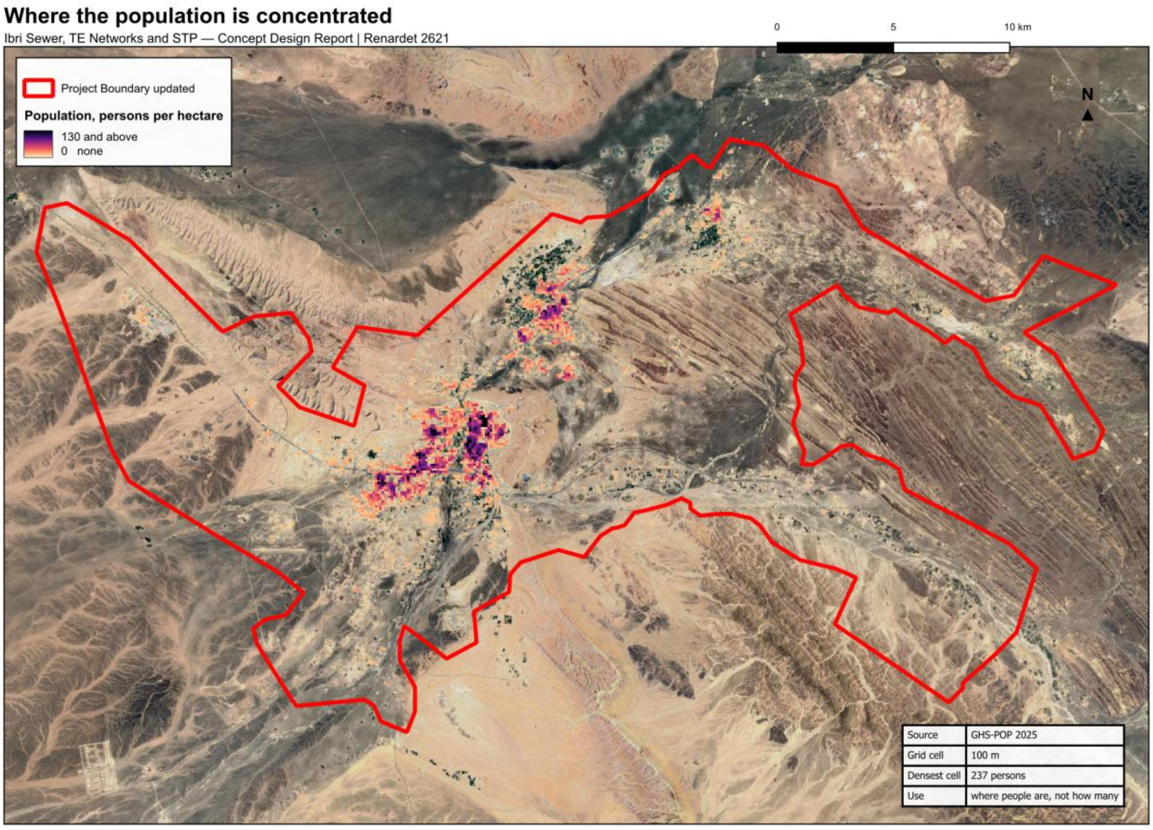


Figure 10 Where the population of the study area lies. Density is concentrated in Ibri town and along the wadi corridors, and falls away sharply beyond them.

14.2 Settlements

Every rule in this Part is applied settlement by settlement: the properties per plot, the occupancy, the share of empty land that becomes housing, the growth rate and the year the land is full are all properties of a settlement, not of the study area. Every plot must therefore belong to exactly one settlement. The settlement boundaries supplied with the Inception Report are polygons drawn around the built cores; they do not touch one another and they leave the land between them unassigned. Each plot has been assigned to the settlement whose boundary contains it, or, where none does, to the nearest, and the twenty-five boundaries

[1] GHS-POP, European Commission Joint Research Centre, 2025 epoch at 100 m resolution, licensed under Creative Commons Attribution 4.0. The grid distributes census counts across built-up surface and is therefore modelled rather than observed.

have been redrawn as a partition of the whole study area, so that every plot lies in one settlement and the boundaries meet without gaps.

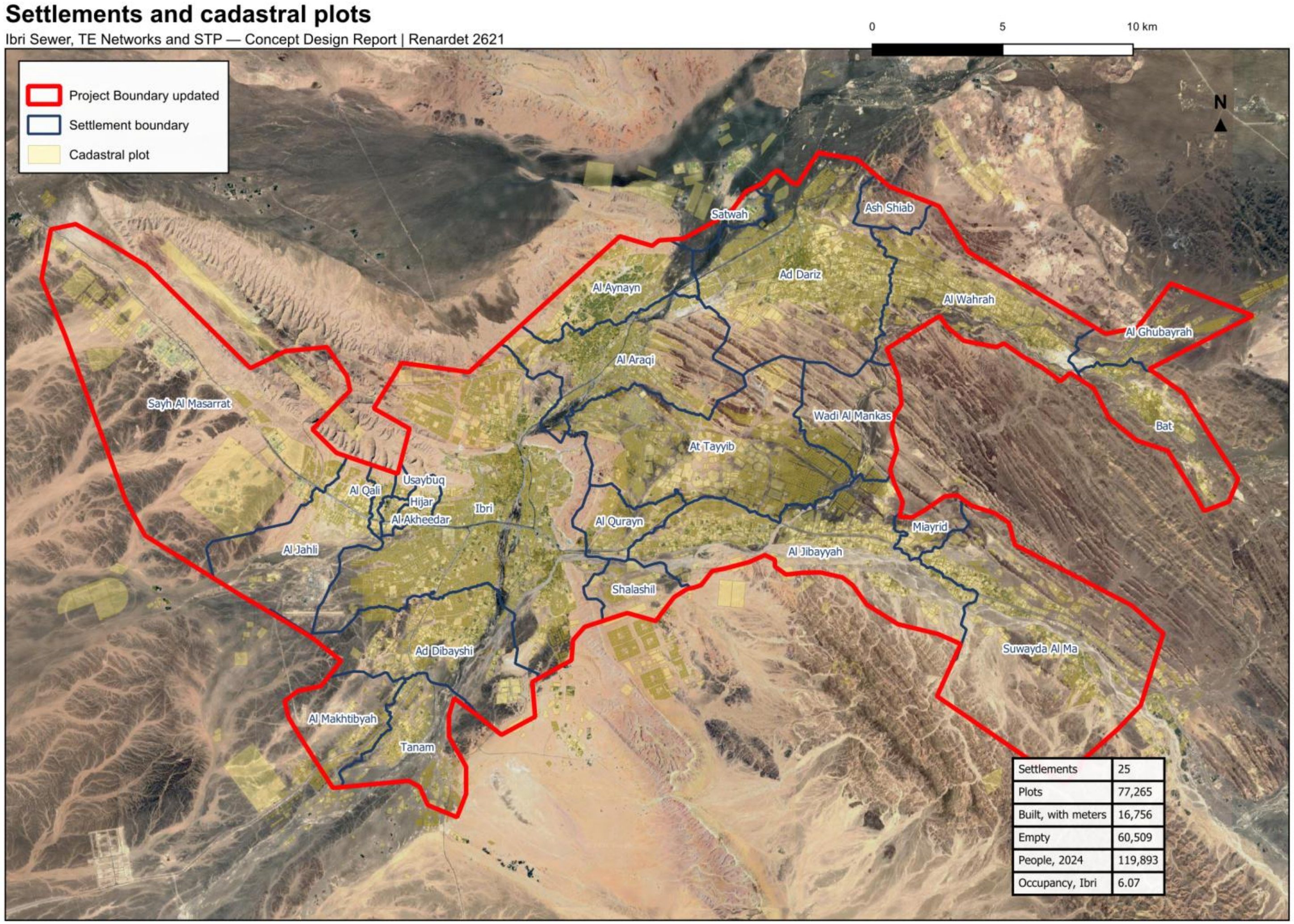


Figure 11   The twenty-five settlements as a partition of the study area, over the cadastral plots. Every plot belongs to one settlement, and the rates used in this Part are set for each.

14.3 Properties per plot

Each domestic electricity meter is one property. Of the 33,971 meters, 30,931 fall inside a plot; 2,914 lie on a road or in a gap between plots and have been assigned to the nearest plot within fifteen metres; 126 lie further from any plot and are listed with their settlement and carry no load in the plot table. The plot table holds 22,559 domestic properties.[1]

The number of properties on a built home plot is measured in each settlement over the plots that are homes and nothing else, excluding the few plots that carry fifteen or more dwelling meters, which are blocks of flats and workers' housing and would distort the ratio. Plots that also carry shops, farms or offices are counted for their people but not for the ratio.

Table 14 Properties per home plot, the principal settlements

Settlement	Domestic properties	Home plots measured	Properties per home plot
Ibri	11,052	5,853	1.33
Ad Dariz	2,439	1,567	1.20
Al Araqi	2,466	1,282	1.32
Al Aynayn	889	560	1.14
Al Wahrah	716	420	1.22
At Tayyib	833	623	1.09
Al Jibayyah	524	291	1.26
Bat	418	239	1.17

Ibri carries 1.33 properties per home plot; the principal settlements lie between 1.1 and 1.4. Four small settlements with institutional housing on a handful of plots return two or more; the thirteen settlements under a thousand people take one property per plot, as Section 14.4 sets out. The full table is given in Appendix A3.[2]

14.4 Occupancy rate

The occupancy rate of each settlement is its population in 2024, the year of the electricity data, divided by the domestic properties counted in it.

$$\mathrm{OR} = \frac{\text{Settlement population in 2024}}{\text{Domestic properties}} \tag{3}$$

Symbol	Meaning	Unit
OR	occupancy rate	persons per property

Two rules bound the derived rate. A floor of 4.0 persons per property: some settlements would otherwise return between one and four, because their properties include the housing of the police headquarters, the college and similar institutions, whose occupants the census does not count as residents; those meters discharge to the sewer whatever the census says, and the floor keeps them in the design. And a cap at the highest rate among the settlements of two thousand or more people, 6.12.[3]

A settlement of fewer than a thousand people in 2024 is treated by one rule for all its rates: an occupancy of 4.0, one property per plot, and a home share of 0.9. It has too few meters to measure on, and a village of that size does not attract second dwellings on a plot. The rule applies to 13 settlements: Al Ghubayrah, Al Qurayn,

[1] The primary, subsidised primary and additional-dwelling tariffs are counted as properties. Shop, government, farm and large-consumer meters carry no people.

[2] 12,443 home plots were measured across the twenty-five settlements.

[3] With the floor and the cap the study area holds 119,893 people against 116,452 in the official series for the same twenty-five settlements; the difference is the institutional housing.

Sayh al Masarrat, Al Jahli, Satwah, Al Akheedar, Al Qali, Shalashil, Al Makhtibyah, Usaybuq, Wadi al Mankas, Ash Shiab, Miayrid.[1]

Table 15 Occupancy rate by settlement

Settlement	Domestic properties	Census population 2024	Rate derived	Rate adopted	People in the design, 2024
Ibri	11,052	67,106	6.07	6.07	67,106
Ad Dariz	2,439	11,850	4.86	4.86	11,850
Al Araqi	2,466	10,696	4.34	4.34	10,696
Al Aynayn	889	4,554	5.12	5.12	4,554
Al Wahrah	716	3,351	4.68	4.68	3,351
At Tayyib	833	3,348	4.02	4.02	3,348
Al Jibayyah	524	2,686	5.13	5.13	2,686
Bat	418	2,557	6.12	6.12	2,557
Ad Dibayshi	681	2,411	3.54	4.00	2,724
Tanam	403	2,116	5.25	5.25	2,116
Suwayda al Ma	301	1,559	5.18	5.18	1,559
Hijar	352	1,033	2.93	4.00	1,408
Al Ghubayrah	161	631	3.92	4.00	644
Al Qurayn	147	549	3.73	4.00	588
Sayh al Masarrat	457	466	1.02	4.00	1,828
Al Jahli	344	415	1.21	4.00	1,376
Satwah	27	263	9.73	4.00	108
Al Akheedar	120	198	1.65	4.00	480
Al Qali	56	191	3.41	4.00	224
Shalashil	38	130	3.42	4.00	152
Al Makhtibyah	91	122	1.34	4.00	364
Usaybuq	21	91	4.33	4.00	84
Wadi al Mankas	8	52	6.49	4.00	32
Ash Shiab	11	47	4.27	4.00	44
Miayrid	4	34	8.49	4.00	16

Ibri returns 6.07 persons per property. Over the study area as a whole the census gives 5.16. Revision 1 of this report adopted a single rate of 5.32 for all settlements; it was the same division over the properties then attributed to a plot, before the meters on roads and in gaps were assigned. The rate per settlement replaces it, and every population and flow in this report follows from the rates in the table.[2]

[1] 4.0 is the rate measured at At Tayyib, 3,300 people, the smallest settlement with a sample worth measuring. The nine settlements of 800 to 5,000 people pool to 4.6.

[2] 116,452 divided by 22,559 properties. Revision 1 divided by 21,889.

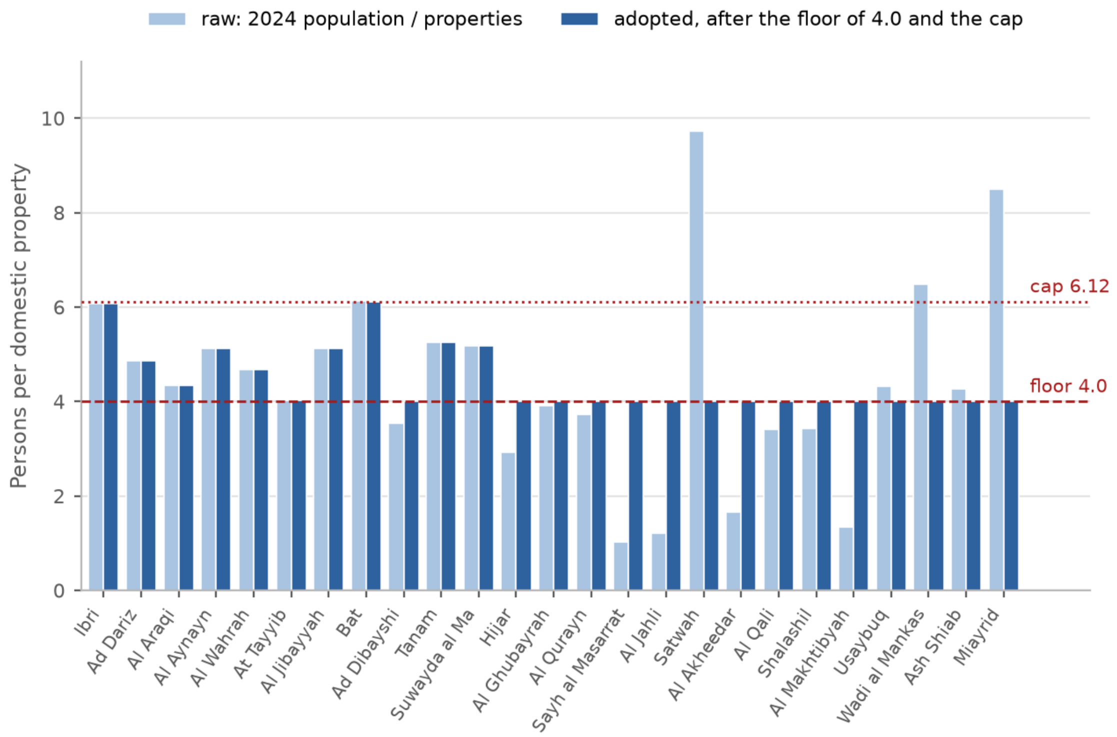


Figure 12 Occupancy by settlement: the derived value and the value adopted after the floor and the cap.

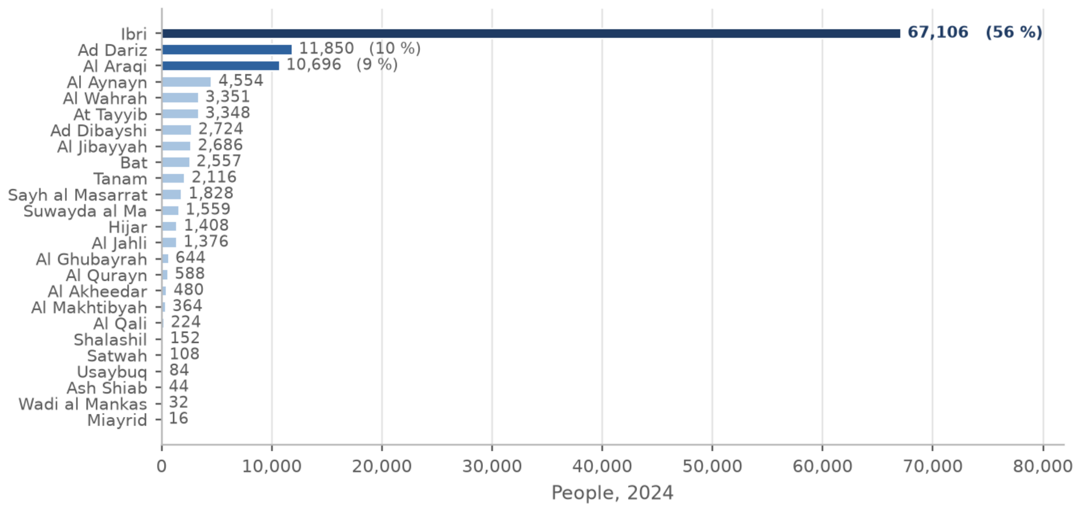


Figure 13 People in each settlement in 2024, from the domestic meters at the settlement's occupancy. Ibri holds 56 per cent of the total.

14.5 Land use

The cadastral layer received in September 2026 holds 77,265 plots. Its land-use field does not carry a government class, records whole districts under codes that mean unclassified, and on the plots that carry meters disagrees with the meters one time in seven. The use of every plot has therefore been derived, in a fixed order, from what is on it.

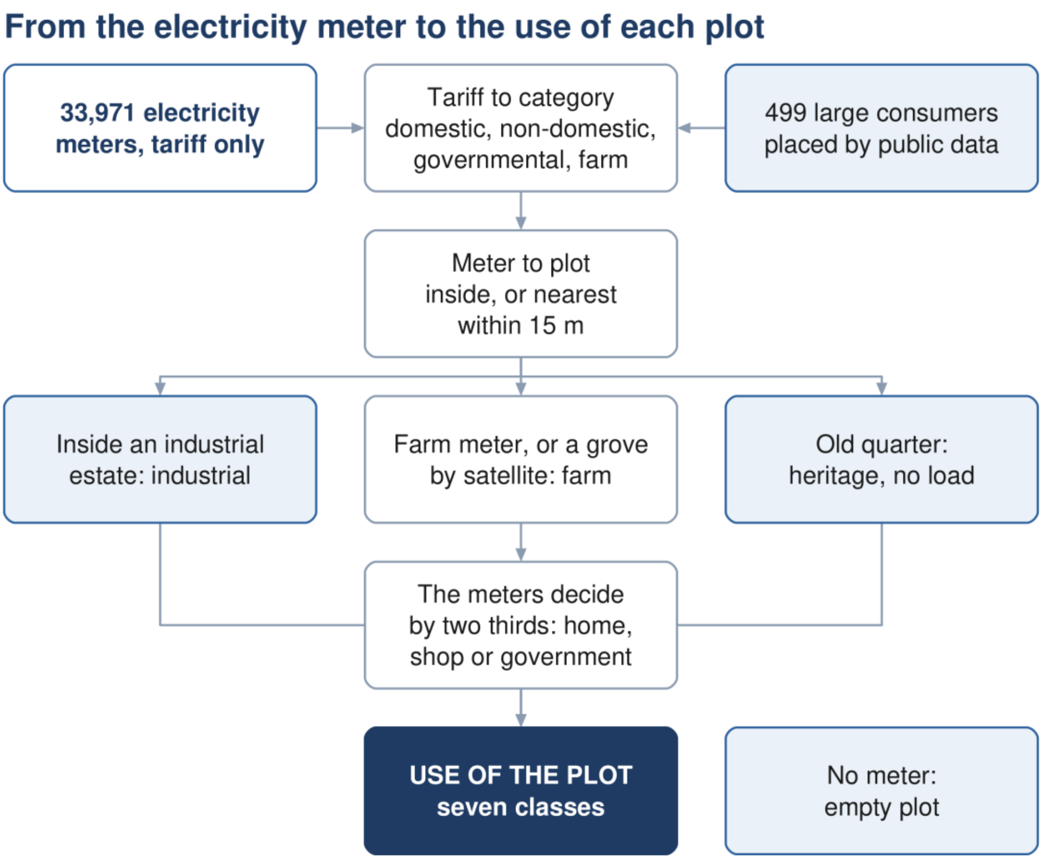


Figure 14 How the use of each plot is derived from the electricity meters, the satellite and the plot itself.

1. a plot inside one of the two industrial estates is industrial;
2. the 177 plots of the old quarter of Ibri, recorded as tourism in the cadastre, are heritage and carry no load;
3. a plot with an agricultural meter is a farm;
4. a plot that the satellite shows as a grove is a farm, unless two thirds of its meters are shops or a majority are government;
5. a plot whose industrial meters equal or outnumber its dwellings is industrial;
6. otherwise the meters decide by proportion: more than two thirds home makes a home; two thirds or more government makes a government plot; two thirds or more shop makes a shop; between those, a home with a shop where the shop meters are at least as many as the government meters, and a government plot where they are fewer;
7. a plot without a meter is empty.

The satellite test uses one cloud-free Sentinel-2 scene of 9 September 2026. Bare ground in the study area returns a vegetation index of 0.08 and a date grove 0.8, and the palms are evergreen, so a single scene separates the two. A plot is a grove where at least 1,000 square metres of it read above 0.30 with a mean above 0.20, or, for a small plot, where it is at least sixty per cent green. The thresholds were set on the 489 plots that carry a farm meter and checked by inspection against the aerial imagery.[1]

Table 16 Use of the built plots that carry a meter, and the heritage quarter

Use	Plots	Basis
Home	12,464	more than two thirds of the meters domestic
Home with a shop	614	mixed, with at least as many shop as government meters
Shop	1,321	two thirds or more of the meters commercial
Government	409	two thirds or more governmental; mixed, with more government than shop meters; or a majority on a green plot
Farm	1,685	a farm meter (489) or a grove by satellite (1,196)
Industrial	263	inside an industrial estate, or industrial meters that match the dwellings
Heritage	177	the old quarter of Ibri; no meter, no load

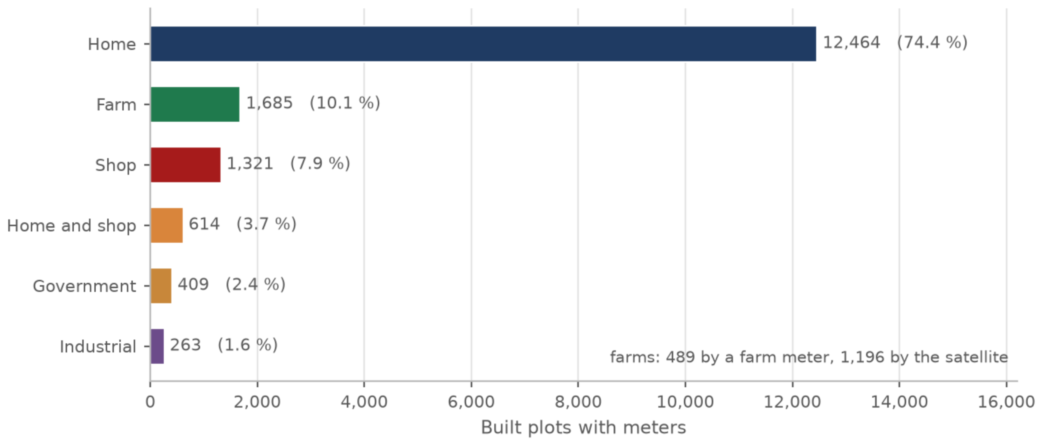


Figure 15 Use of the built, metered plots as derived. Homes are 74 per cent of the plots; farms, 10 per cent, are found by the meter or by the satellite.

[1] Sentinel-2 Level 2A, tile 40QDL, red and near-infrared bands at 10 m, normalised difference vegetation index. 38 per cent of the plots with a farm meter fail the grove test: pump sites without a crop. That is why the meter alone under-counts farms.

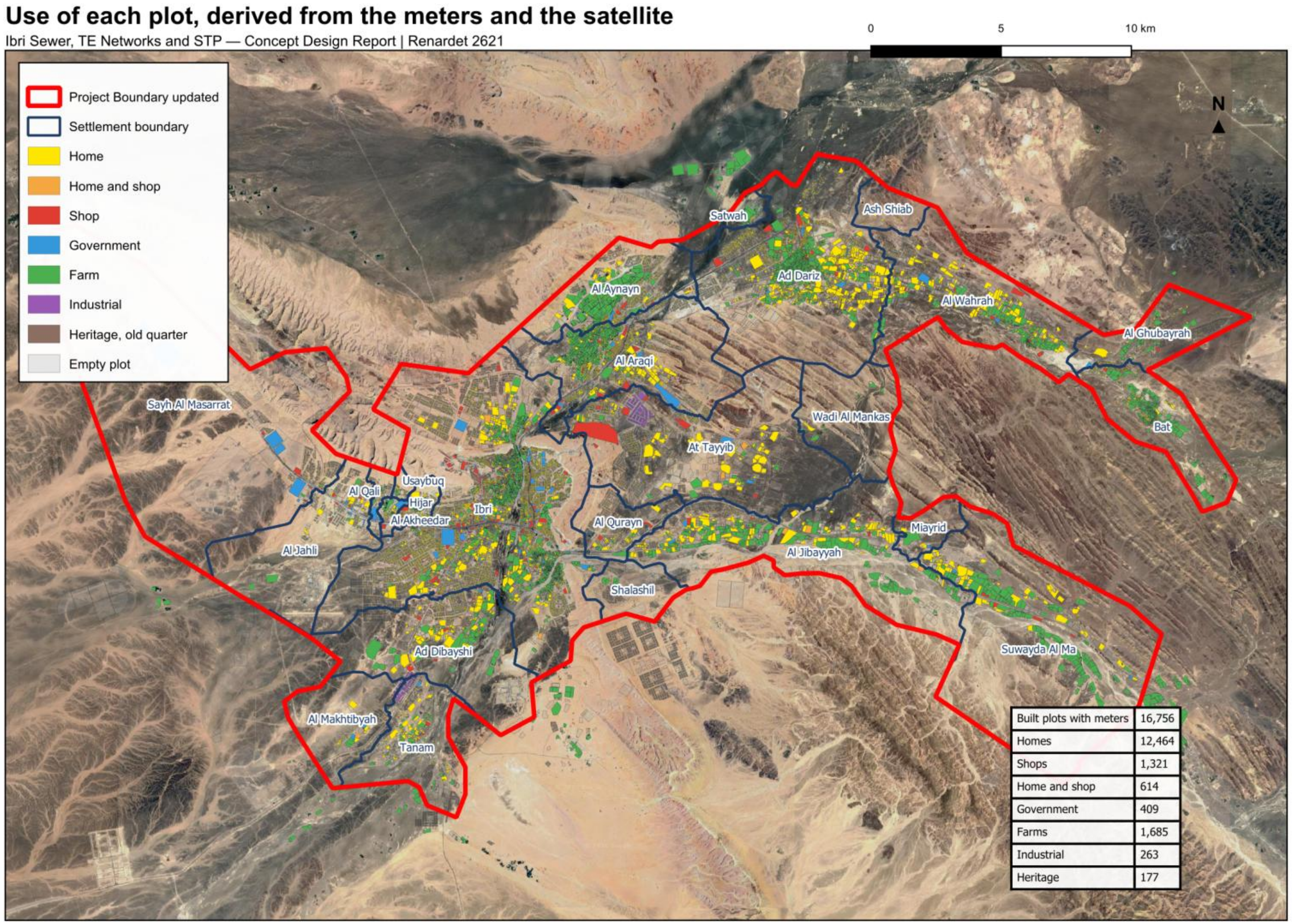


Figure 16 Use of each plot as derived from the meters and the satellite. Empty plots are shown in grey.

14.6 Capacity of the empty land

Of the 77,265 plots, 60,509 carry no meter. Not all of them will become homes. A future home plot must look like the built ones: between 200 and 1,000 square metres, compact, and not a strip. Slivers below 200 square metres, odd shapes and everything above 1,000 square metres are left out of the count, as are groves, industrial plots and the heritage quarter. 46,109 empty plots pass that test.[1]

A district also needs mosques, shops and offices, so not every home-shaped plot becomes a home. The share that does is measured in each settlement over its built, metered, home-shaped plots: pure homes divided by all. Ibri returns 0.87; among the settlements of a thousand people or more the measured share runs from 0.67 in Tanam, with its workshops, to 0.94 in Al Aynayn.

The capacity of a settlement is then its home-shaped empty plots, multiplied by its home share, by its properties per home plot and by its occupancy rate.

$$\text{Capacity} = \text{Empty home-shaped plots} \times \text{Home share} \times \text{Properties per plot} \times \text{OR} \quad (4)$$

Symbol	Meaning	Unit
Capacity	people the empty land of the settlement can hold	persons
Home share	share of home-shaped plots that become homes	—
OR	occupancy rate of the settlement	persons per property

The empty land of the study area holds 229,136 people on this basis, against 119,893 living in it in 2024.[2]

[1] Compactness is the plot's area against that of its smallest enclosing rectangle, not less than 0.6; the rectangle's sides may not differ by more than three to one. Built home plots have a median area of 717 square metres and a median compactness of 0.98. Shapes above 2,000 square metres are whole future districts drawn as one plot and take nothing until they are subdivided.

[2] The thirteen settlements under a thousand people take a home share of 0.9 and one property per plot, as Section 14.4 sets out.

14.7 Growth and the year the land is full

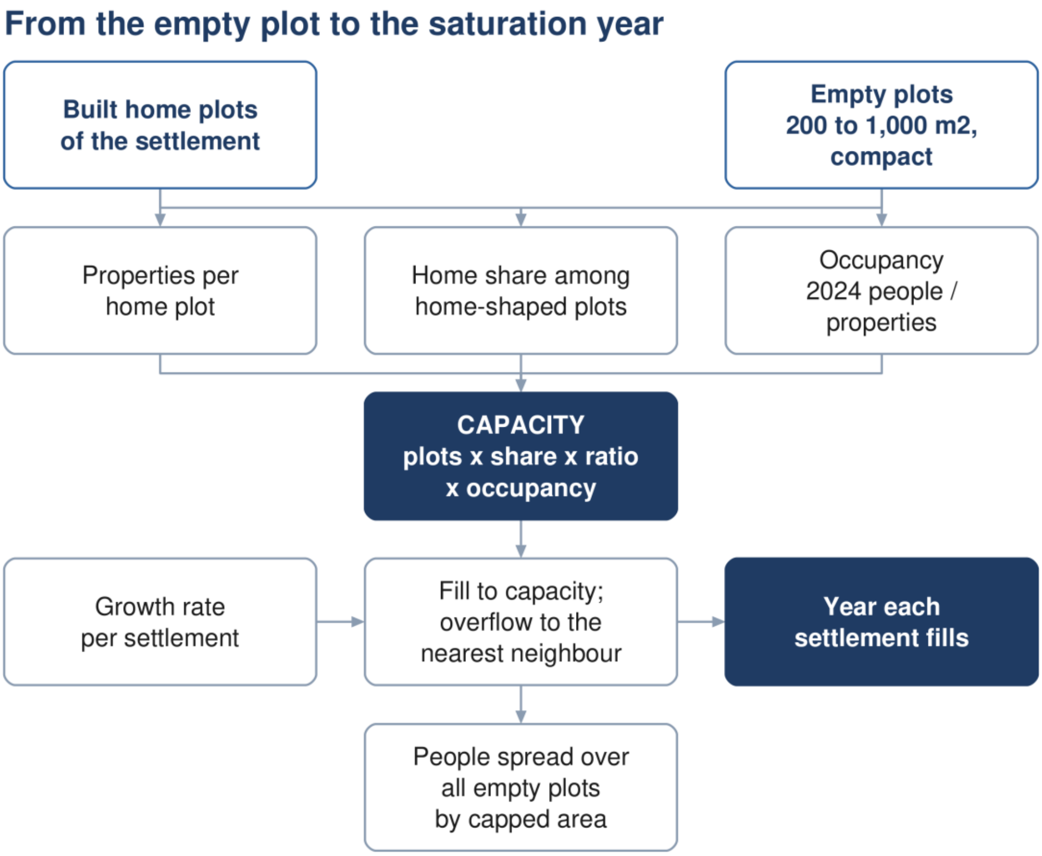


Figure 17 From the empty plot to the year each settlement fills.

The growth series

The population series of the Inception Report is used for its growth rates. It is built in three parts. To 2040 it is the National Centre for Statistics and Information forecast for the Wilayat of Ibri, Omani and expatriate population separately. From 2041 to 2050 it is an extrapolation of that forecast, documented in the technical note on population issued with the Inception Report. From 2051 it rises from 2.2 per cent a year to 2.40 by 2058 and stays at 2.40 to 2100, the planning horizon instructed by Nama Water Services. The wilayat population is then split between settlements in fixed shares from the census, so every settlement carries the same annual rate. Over 2024 to 2030 that rate averages 2.6 per cent a year, over the 2030s 2.5, over the 2040s 2.0, over the 2050s 2.3 and 2.4 from 2060. The series is not a saturation figure: the land fills long before 2100.[1]

[1] Inception Report Revision 0, Section 6, and the demand workbook issued with it, sheets Pop_Wilayat and Project Pop Settlements. Wilayat population 183,564 in 2024; the twenty-five settlements are 63.4 per cent of it in every year. The workbook total for the settlements is 691,264 in 2100.

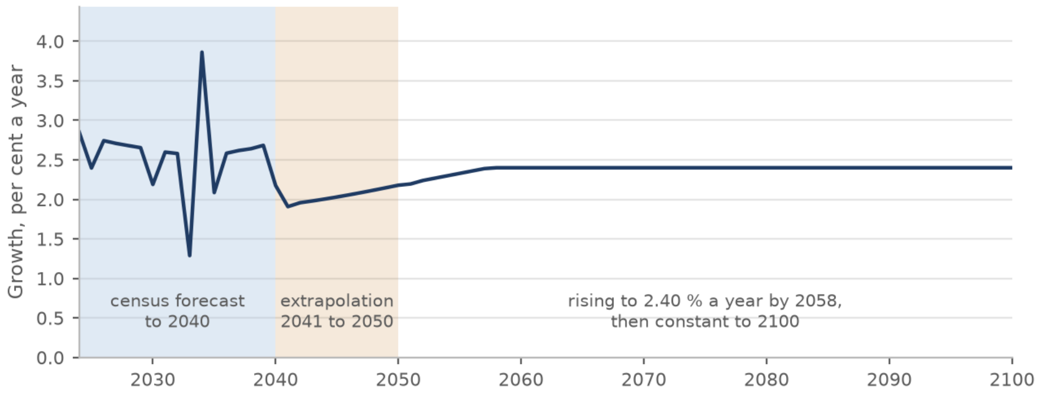


Figure 18 The annual growth rate of the population series, by year: the census forecast to 2040, the extrapolation to 2050, and the rise to a constant 2.40 per cent beyond.

Filling the land

Each settlement grows at the rate of the series, applied to its counted population, and its growth fills its empty plots together and in proportion. When a settlement is full, its further growth moves to a neighbour. Ibri's overflow goes in parallel to Al Araqi, Al Qurayn and Shalashil, in the proportions seventy, twenty and ten per cent, then to Ad Dariz when those are full, then to the nearest settlement with room. At Tayyib's overflow goes first to Miayrid, the small settlement beside it. Every other settlement overflows to its nearest neighbour with room. The year a settlement's empty plots are full is its saturation year; the study area is saturated when the last of them fills.[1]

The rule sets the order; the land sets the result. Al Araqi fills in 2057, a year after Ibri, so it takes only 163 of Ibri's people, and most of Ibri's overflow passes on: the largest route is to At Tayyib, 22,078 people, and 14 settlements take some of Ibri's growth in all.

Table 17 Population at five-year intervals to saturation

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Saturation year	Saturation population
Ibri	67,106	68,713	78,099	88,272	100,066	110,349	122,535	137,198				2056	140,405
Ad Dariz	11,850	12,134	13,791	15,587	17,670	19,486	21,638	24,206	27,350			2062	28,704
Al Araqi	10,696	10,952	12,448	14,069	15,949	17,588	19,531	21,849				2057	22,528
Al Aynayn	4,554	4,663	5,300	5,990	6,791	7,489	8,316	9,303	12,013			2060	12,013
Al Wahrah	3,351	3,431	3,900	4,408	4,997	5,511	6,119	6,845	7,703	8,684	10,460	2070	10,460
At Tayyib	3,348	3,428	3,897	4,404	4,993	5,506	6,114	6,839	8,130	23,972		2068	38,023
Ad Dibayshi	2,724	2,789	3,170	3,583	4,062	4,479	4,974	5,564	6,262			2063	10,644
Al Jibayyah	2,686	2,750	3,126	3,533	4,005	4,416	4,904	5,486	6,174	6,951	13,992	2070	13,992
Bat	2,557	2,618	2,976	3,363	3,813	4,204	4,669					2052	4,783
Tanam	2,116	2,167	2,463	2,784	3,156	3,480	3,864	4,323	4,865	5,961		2069	8,895
Sayh al Masarrat	1,828	1,872	2,127	2,405	2,726	3,006	3,338	3,734	4,202	4,786		2069	6,812
Suwayda al Ma	1,559	1,596	1,814	2,050	2,324	2,563	2,846	3,184	3,583	4,034	7,928	2070	7,928
Hijar	1,408	1,442	1,639	1,852	2,100	2,315	2,571					2053	2,696
Al Jahli	1,376	1,409	1,601	1,810	2,052	2,263	2,513	2,831	3,746			2063	5,060
Al Ghubayrah	644	659	750	847	960	1,059	1,176	1,756	2,575	3,501	4,584	2070	4,584
Al Qurayn	588	602	684	773	877	967	1,074	1,201	7,176			2060	7,176
Al Akheedar	480	492	559	631								2036	640
Al Makhtibyah	364	373	424	479	543	599	665	744	837	1,522		2069	4,656
Al Qali	224	229	261	295	410	518	646					2054	788
Shalashil	152	156	177	200	227	250	278	311	8,335			2061	11,852
Satwah	108	111	126	142	161	178	197	221	392			2060	392
Usaybuq	84	86	98	110	125	138	153					2054	224
Ash Shiab	44	45	51	58	66	72	80	90				2056	92

[1] Inception Report demand workbook, sheet Project Pop Settlements, 2024 to 2100. The series is used for its growth rates only. Its total for 2100 is not a target and is not a saturation figure: the land is full before it is reached. The overflow routes reflect local knowledge of where Ibri's growth is taking place.

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Saturation year	Saturation population
Wadi al Mankas	32	33	37	42	48	53	58	65	74	2,859		2068	5,052
Miayrid	16	16	19	21	24	26	29	33	37	41	632	2070	632
Total	119,893	122,766	139,535	157,710	178,781	197,153	218,926	244,914	275,608	310,307	349,029	2070	349,029

Ibri fills in 2056 and Al Araqi in 2057, Al Qurayn in 2060, Shalashil in 2061 and Ad Dariz in 2062. The last settlement fills in 2070, which is the saturation year of the study area, with 349,029 people. A cell in the table is blank once the settlement is full.[1]

The routes the overflow takes, and when, are set out below for every route carrying more than a thousand people; the full list is in Appendix A5, and the year each settlement would fill on its own growth alone is in Appendix A6.

Table 18 The main overflow routes

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Ibri	At Tayyib	22,078	2056	2060	2068
Ibri	Shalashil	11,471	2056	2056	2061
Ibri	Al Qurayn	5,856	2056	2056	2060
Ad Dariz	Wadi al Mankas	4,965	2062	2062	2068
Al Araqi	At Tayyib	4,873	2057	2060	2068
Ibri	Ad Dibayshi	4,078	2056	2062	2063
Ibri	Al Jibayyah	3,927	2056	2068	2070
Bat	Al Ghubayrah	2,493	2052	2052	2070
Ibri	Al Makhtibyah	2,441	2056	2063	2069
Ibri	Tanam	2,010	2056	2063	2069
Al Araqi	Al Aynayn	1,790	2057	2057	2060
Al Aynayn	At Tayyib	1,639	2060	2060	2068
Al Araqi	Al Jibayyah	1,410	2057	2068	2070
Ibri	Sayh al Masarrat	1,225	2056	2063	2069
Ad Dibayshi	Tanam	1,004	2063	2063	2069

The overflow matters. On its own growth alone, 7 of the twenty-five settlements would not fill before 2100; with the overflow, every settlement fills by 2070. A network in those settlements sized on their own growth would be sized for land that never fills; sized on the overflow, it serves the plots that Ibri's growth actually occupies.[2]

[1] Totals are rounded from the unrounded values and may differ by one from the sum of the rows. A settlement's population after its saturation year is its capacity plus what it held in 2024; its growth beyond that is housed elsewhere or not at all.

[2] The settlements that never fill on their own growth: At Tayyib, Al Ghubayrah, Al Qurayn, Al Makhtibyah, Shalashil, Wadi al Mankas, Miayrid.

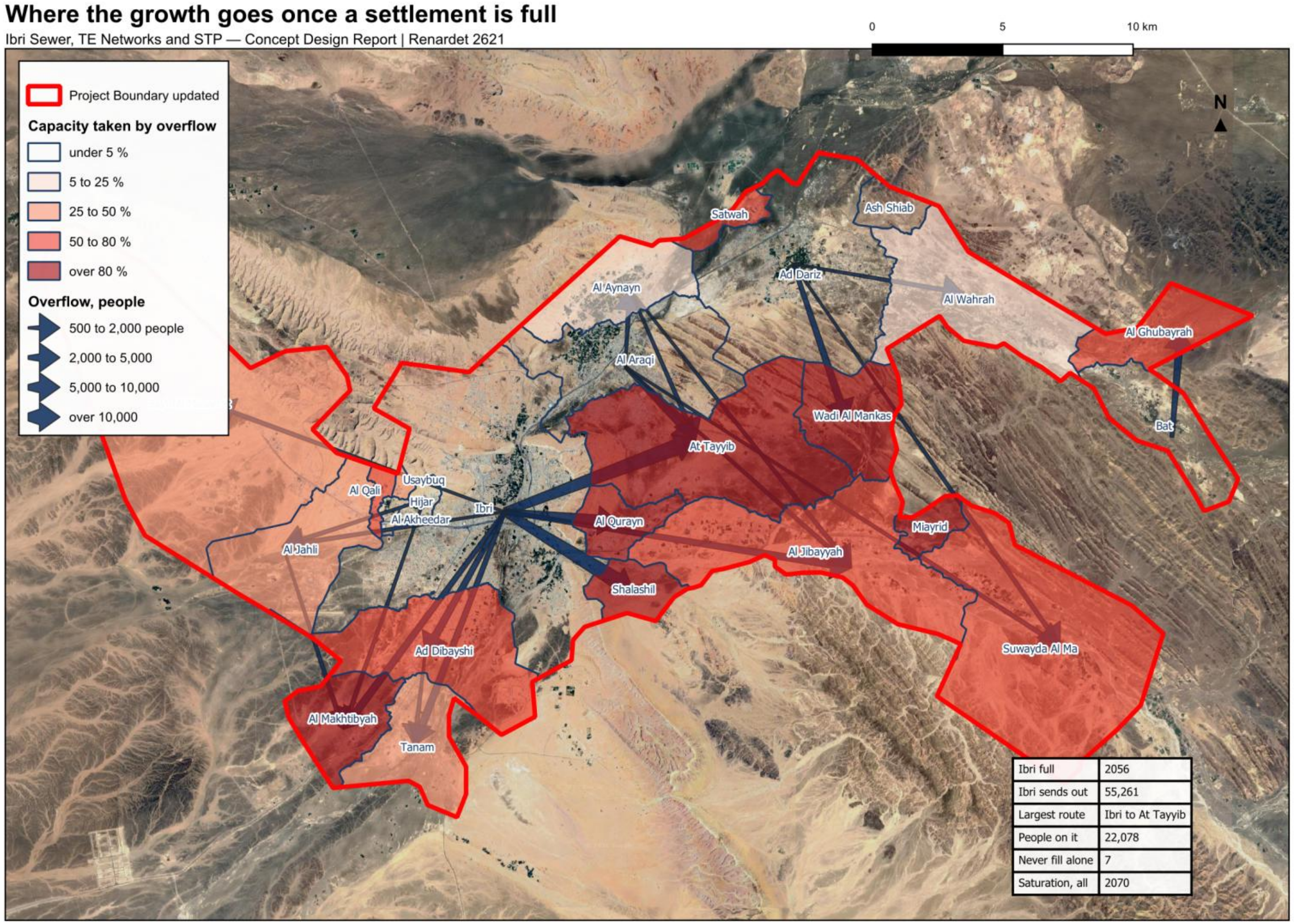


Figure 19 Where the growth goes once a settlement is full. Arrows show the routes carrying five hundred people or more at saturation; the shading shows how much of each settlement's capacity is taken by its neighbours' overflow.

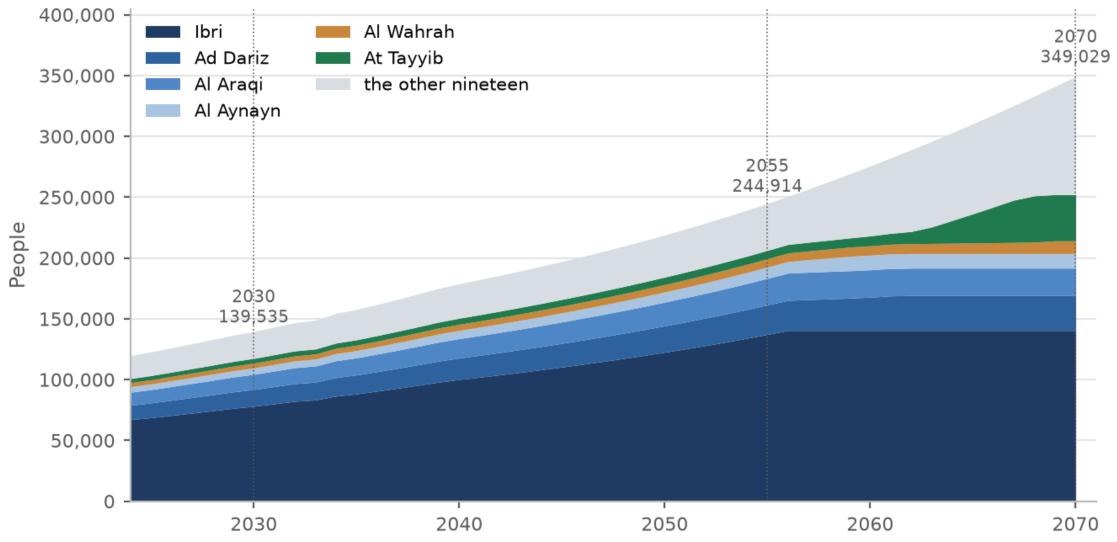


Figure 20 People by year to saturation, the six largest settlements shown separately. The flat top of each band is the year that settlement fills.

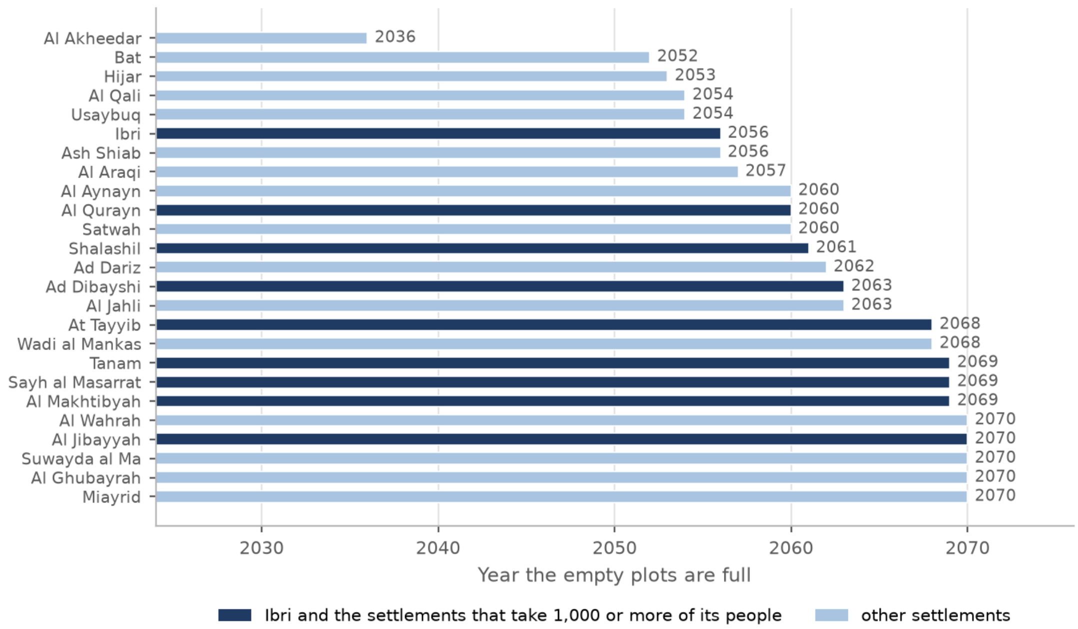


Figure 21 The year each settlement's empty plots are full. Ibri and the settlements that take a thousand or more of its people are marked.

14.8 Where the growth is placed

For the network every empty plot matters, not only the ones counted as future homes, because the sewer that serves a street serves every plot on it. The people a settlement houses in a given year are therefore spread over all its empty plots up to 2,000 square metres, in proportion to plot area capped at 1,000 square metres. A sliver takes a sliver's share and a large plot takes no more than a full-sized home plot. The sum over the plots equals the settlement's housed population by construction, so the network total and the treatment plant total are the same figure.

15 Wastewater generation and design flows

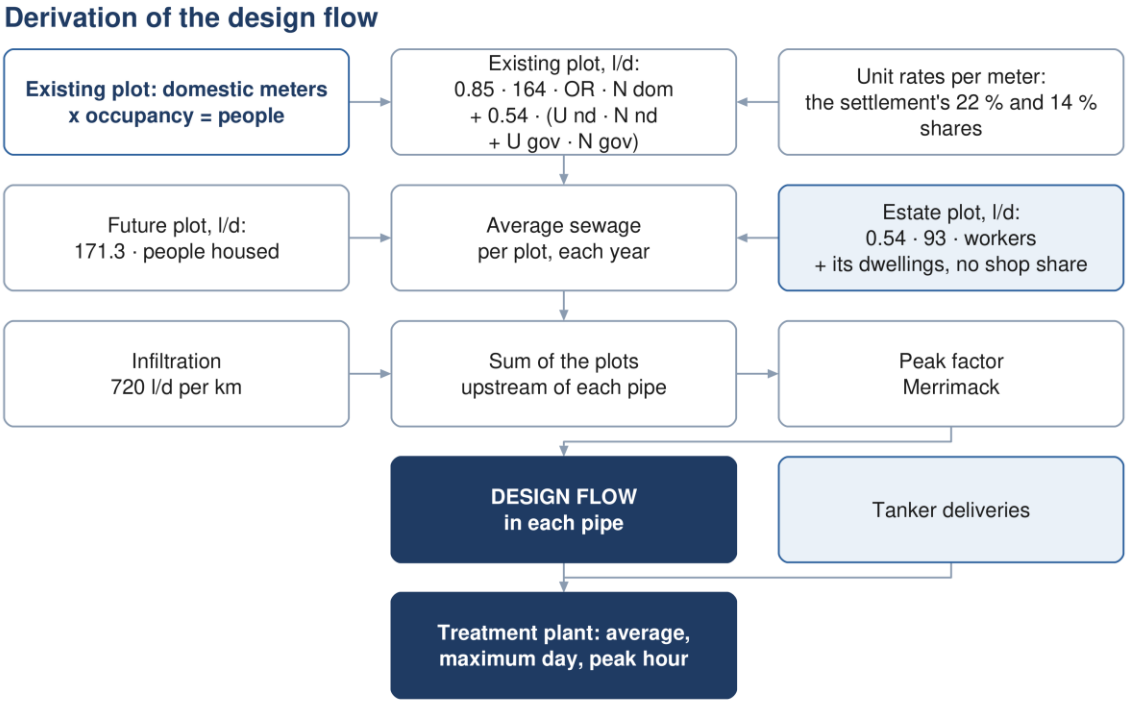


Figure 22 Derivation of the design flow, from the plot to the pipe and the treatment plant.

15.1 Water demand

Wastewater generation is derived from water demand. Four components are carried: domestic, non-domestic, governmental and special consumption. A household supplied by tanker is metered for electricity like any other and carries the domestic rate; the tanker volumes themselves are a data request and no separate term is carried (Section 10.1).

$$Q_{\mathrm{dom}} = \mathrm{Population} \times \mathrm{LPCD} \tag{5}$$

Symbol	Meaning	Unit
Q dom	domestic water demand	l/d
LPCD	unit consumption, 164 l/c/d for Adh Dhahirah	l/c/d

Non-domestic and governmental demand are established as proportions of domestic demand.

$$Q_{\mathrm{ND}} = 0.22 Q_{\mathrm{dom}} \tag{6}$$

$$Q_{\mathrm{gov}} = 0.14 Q_{\mathrm{dom}} \tag{7}$$

The proportions are the values published for the Governorate of Adh Dhahirah. They represent the non-domestic and governmental volumes recorded in the governorate water balance expressed against domestic consumption. Per person they are 36 and 23 litres a day, and the three are applied as separate streams, never combined into one rate. Each settlement's non-domestic and governmental water is placed on its shop and government meters in proportion to their number.[1]

[1] PAM-GUD-201, Table 11, page 60. The column headings describe the ratios as distributed, being governorate volumes recorded between 2021 and 2023.

15.2 Special consumption: the industrial estates

Two industrial estates lie inside the study area, at Al Tayyeb in the north-east of the town and at Tanam beside the treatment plant. Their meters are recorded on the commercial tariff, which is why an earlier reading of the data found no industrial estate. The guideline places such estates outside the population ratios, and they are treated as special consumption: a workforce of 4,500 at Al Tayyeb and 1,800 at Tanam, spread over their industrial plots in proportion to area, at the dry-industry rate of 93 litres per employee per day. Their meters take no share of the settlement's non-domestic or governmental water: an estate plot carries its workers, and the few dwellings on the estates carry the domestic rate like any other. The workforce is an assumption from the number of workshop meters and is to be replaced by the estates' own records. Section 16 describes the estates and the other identified projects.[1]

15.3 Return to the sewer

Not all supplied water reaches the sewer. The proportion that does is applied by category.

$$Q_{\mathrm{ww}} = 0.85Q_{\mathrm{dom+tanker}} + 0.54Q_{\mathrm{ND+gov+special}} \qquad (8)$$

Symbol	Meaning	Unit
Q ww	wastewater generated, before infiltration	m³/d
Q dom+tanker	domestic demand; tanker supply added when its records are received	m³/d
Q ND+gov+special	non-domestic, governmental and special demand	m³/d

Water supplied by tanker returns to the sewer at the same proportion as piped supply. The domestic rate of 164 litres a day measures network water only, so tanker supply is not inside the domestic stream; it is added when the filling-station records are received (Section 10.1).[2]

Agricultural meters supply irrigation pumps and are taken to return nothing, and the estates' dry-industry demand is taken to return 54 per cent, as non-domestic supply does. Both are rules of this design, not of the guideline.

15.4 The flow from each plot

Applied plot by plot, the rules above give every plot an average daily sewage flow for 2024. An existing plot is built from its meters and the rates of its settlement:

$$Q_{\mathrm{plot}} = 0.85 \times 164 \times OR \times N_{\mathrm{dom}} + 0.54 \times \left(U_{\mathrm{nd}}N_{\mathrm{nd}} + U_{\mathrm{gov}}N_{\mathrm{gov}} + 93N_{\mathrm{w}}\right) \qquad (9)$$

Symbol	Meaning	Unit
Q plot	average sewage flow of the plot in 2024	l/d
OR	occupancy adopted for the settlement (Table 15)	persons per property
N dom	domestic meters on the plot, one property each	—
U nd	non-domestic water per shop meter in the settlement	l/d per meter
N nd	non-domestic meters on the plot; none counted on an estate plot	—
U gov	governmental water per government meter in the settlement	l/d per meter
N gov	government meters on the plot; none counted on an estate plot	—
N w	workers on an industrial-estate plot	—

The unit rates are the settlement's non-domestic and governmental shares, divided equally among the meters that carry them:

[1] PAM-GUD-201, Section 7.3.1, page 59, and Table 12, page 61, dry industry. Al Tayyeb: 94 hectares, 202 industrial plots, 1,035 meters. Tanam: 61 hectares, 105 plots, 409 meters. About five meters per plot and three to six workers per workshop give the assumed figures.

[2] PAM-GUD-201, Table 19, page 71, which gives a single discharge ratio for domestic and tanker supply.

$$U_{\mathrm{nd}} = \frac{0.22 \times 164 \times P_s}{N_{\mathrm{nd,s}}} \tag{10}$$

$$U_{\mathrm{gov}} = \frac{0.14 \times 164 \times P_s}{N_{\mathrm{gov,s}}} \tag{11}$$

Symbol	Meaning	Unit
P_s	people of the settlement in 2024	—
$N_{nd,s}$	shop meters of the settlement outside the industrial estates	—
$N_{gov,s}$	government meters of the settlement outside the industrial estates	—

The estates' meters take no share of the pool; an estate plot carries its workers at 93 litres a day, and the dwellings on it, if any, at the domestic rate. A settlement with no shop or government meter outside the estates keeps that share on its dwellings, in proportion to their people. A future plot carries the people it houses in the year at 171 litres a day each, all three streams together, since where its shops will stand is not known.[1]

Worked for an illustrative plot in Ibri with four domestic, two commercial and one government meter, at the settlement's occupancy of 6.07 and its unit rates of 510 and 3,342 litres a day per meter:

Table 19 The flow of one plot, worked from its meters

Stream and meters	Water, l/d	Return	Sewage, l/d
Domestic: 4 meters × 6.07 persons × 164 l/d	3,983	0.85	3,386
Non-domestic: 2 meters × 510 l/d	1,020	0.54	551
Governmental: 1 meter × 3,342 l/d	3,342	0.54	1,805
Plot	8,345		5,741 = 5.74 m³/d

The shares and the unit rates of every settlement are set out below. The sum of the plots equals the settlement in every column.[2]

Table 20 Water shares and unit rates by settlement, 2024

Settlement	People	Domestic m³/d	Non-domestic share m³/d	Governmental share m³/d	Shop meters	Government meters	l/d per shop meter	l/d per government meter	Estates m³/d	Sewage m³/d
Ibri	67,106	11,005	2,421	1,541	4,747	461	510	3,342	–	11,494
Ad Dariz	11,850	1,943	428	272	820	82	521	3,318	–	2,030
Al Araqi	10,696	1,754	386	246	1,018	96	379	2,558	–	1,832
Al Aynayn	4,554	747	164	105	204	26	805	4,022	–	780
Al Wahrah	3,351	550	121	77	132	23	916	3,345	–	574
At Tayyib	3,348	549	121	77	179	19	675	4,046	419	799
Al Jibayyah	2,686	440	97	62	96	20	1,009	3,083	–	460
Bat	2,557	419	92	59	51	26	1,809	2,258	–	438
Ad Dibayshi	2,724	447	98	63	236	17	416	3,679	27	481
Tanam	2,116	347	76	49	124	32	616	1,518	105	419
Suwayda al Ma	1,559	256	56	36	34	13	1,654	2,753	–	267
Hijar	1,408	231	51	32	63	4	806	8,082	–	241
Al Ghubayrah	644	106	23	15	105	4	221	3,697	–	110
Al Qurayn	588	96	21	14	14	7	1,515	1,929	–	101
Sayh al Masarrat	1,828	300	66	42	128	99	515	424	–	313
Al Jahli	1,376	226	50	32	143	87	347	363	–	236
Satwah	108	18	4	2	5	0	779	on dwellings	–	19
Al Akheedar	480	79	17	11	69	6	251	1,837	–	82
Al Qali	224	37	8	5	5	22	1,616	234	–	38

[1] 164 × 0.85 + 36 × 0.54 + 23 × 0.54 = 171.3 litres per person per day.

[2] The Tanam estate's plots lie across Tanam, Ad Dibayshi and Al Makhtibyah in the settlement partition, so its workforce appears under all three. A settlement with no shop or government meter outside the estates keeps that share on its dwellings. Two plots outside the estates carry a special meter each, one on the industrial tariff and one large consumer identified as industrial; they are loaded through their shop meters, their own consumption being unknown. Totals are rounded from the unrounded values and may differ by one from the sum of the rows.

Settlement	People	Domestic m³/d	Non-domestic share m³/d	Governmental share m³/d	Shop meters	Government meters	l/d per shop meter	l/d per government meter	Estates m³/d	Sewage m³/d
Shalashil	152	25	5	3	2	1	2,742	3,490	–	26
Al Makhtibyah	364	60	13	8	45	4	292	2,089	36	82
Usaybuq	84	14	3	2	0	1	on dwellings	1,929	–	14
Wadi al Mankas	32	5	1	1	0	0	on dwellings	on dwellings	–	5
Ash Shiab	44	7	2	1	0	1	on dwellings	1,010	–	8
Miayrid	16	3	1	0	4	2	144	184	–	3
Total	119,893	19,663	4,326	2,753	8,224	1,053	526	2,614	586	20,852

In 2024 the study area generates 20,852 cubic metres a day, of which 16,713 is domestic, 2,336 non-domestic, 1,486 governmental and 316 from the two estates.[1]

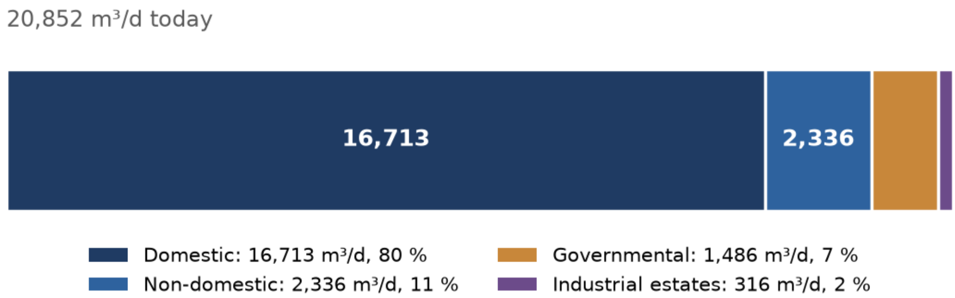


Figure 23 Average sewage flow in 2024 by stream.

15.5 Infiltration

For newly constructed networks an allowance of 720 litres per day per kilometre of sewer is included. It is a property of the pipe, not of the plot, and is added along each run. Infiltration from stormwater is not considered.[2]

15.6 Peak flow

Peak flow is a property of the flow accumulated in a pipe, not of a plot. For catchments of more than one hundred properties the peak daily flow is established by the Merrimack formula, applied to the sum of the plots upstream of each pipe. For a head of one hundred properties or fewer the guideline's alternative, the Peltier formula, is applied.

$$Q_{\text{pdf}} = 2.65 Q_{\text{adf}}{}^{0.879} \tag{12}$$

Symbol	Meaning	Unit
Q pdf	peak daily flow	Ml/d
Q adf	average daily flow	Ml/d

Both flows are expressed in megalitres per day. The peaking factor is the ratio of the two. At the treatment plant the average, the maximum day and the peak hour are taken from the same accumulated flow, with the design margin added and, once their records are received, the tanker deliveries.[3]

[1] The plot layer delivered with this report carries the meters, the people, the unit rates, the water and the sewage of every plot by stream, and the population and flow for 2030, 2055 and the saturation year; the settlement layer carries the same at settlement level with the five-year steps (Appendix A4).

[2] PAM-GUD-201, Section 7.4.3, page 72.

[3] PAM-GUD-201, Section 7.4.2, pages 71 and 72. The Peltier alternative is 1.5 + 1/√Q, with the average flow in litres per second. The guideline states that the formula is to be used for a catchment or sub-catchment having over 100 properties, and recommends that the hourly peak factor should not exceed 5.0.

15.7 Projection through the design period

Flows are established annually and reported at five-year intervals to saturation, as the Terms of Reference require. The study area generates 20,852 cubic metres a day in 2024, 24,216 in 2030, 42,266 in 2055 and 60,099 at saturation in 2070. The network is sized on the saturation flow; the treatment plant is staged on the series.[1]

Table 21 Average sewage flow at five-year intervals to saturation, m³/d

Settlement	2024	2025	2030	2035	2040	2045	2050	2055	2060	2065	2070	Saturation year	Saturation flow
Ibri	11,494	11,769	13,377	15,119	17,140	18,901	20,988	23,500				2056	24,049
Ad Dariz	2,030	2,078	2,362	2,670	3,027	3,338	3,706	4,146	4,685			2062	4,917
Al Araqi	1,832	1,876	2,132	2,410	2,732	3,013	3,345	3,742				2057	3,859
At Tayyib	800	813	893	980	1,081	1,169	1,273	1,398	1,618	4,332		2068	6,739
Al Aynayn	780	799	908	1,026	1,163	1,283	1,424	1,593	2,058			2060	2,058
Al Wahrah	574	588	668	755	856	944	1,048	1,173	1,320	1,487	1,792	2070	1,792
Ad Dibayshi	481	492	557	628	710	782	866	967	1,087			2063	1,838
Al Jibayyah	460	471	535	605	686	756	840	940	1,057	1,191	2,397	2070	2,397
Bat	438	448	510	576	653	720	800					2052	819
Tanam	419	428	479	534	597	653	719	797	890	1,078		2069	1,580
Sayh al Masarrat	313	321	364	412	467	515	572	640	720	820		2069	1,167
Suwayda al Ma	267	273	311	351	398	439	488	545	614	691	1,358	2070	1,358
Hijar	241	247	281	317	360	397	440					2053	462
Al Jahli	236	241	274	310	351	388	430	485	642			2063	867
Al Ghubayrah	110	113	128	145	165	181	201	301	441	600	785	2070	785
Al Qurayn	101	103	117	133	150	166	184	206	1,229			2060	1,229
Al Akheedar	82	84	96	108								2036	110
Al Makhtibyah	82	83	92	101	112	122	133	147	163	280		2069	817
Al Qali	38	39	45	51	70	89	111					2054	135
Shalashil	26	27	30	34	39	43	48	53	1,428			2061	2,030
Satwah	19	19	22	24	28	30	34	38	67			2060	67
Usaybuq	14	15	17	19	22	24	26					2054	38
Ash Shiab	8	8	9	10	11	12	14	15				2056	16
Wadi al Mankas	6	6	6	7	8	9	10	11	13	490		2068	865
Miayrid	3	3	3	4	4	5	5	6	6	7	108	2070	108
Total	20,852	21,344	24,216	27,329	30,938	34,085	37,815	42,266	47,523	53,466	60,099	2070	60,099

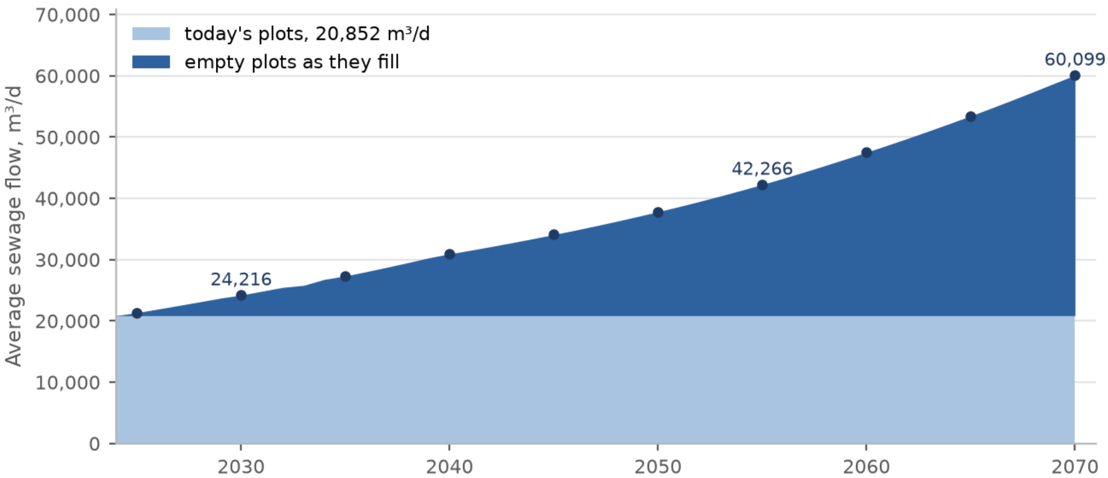


Figure 24 Average sewage flow of the study area by year, from today's plots and from the empty plots as they fill. The points mark the five-year reporting intervals.

[1] Average daily flows before infiltration and before the plant margin. The ultimate flow of about 49,700 cubic metres a day stated in the Inception Report was built on the connected population and a single occupancy rate; the figures here supersede it.

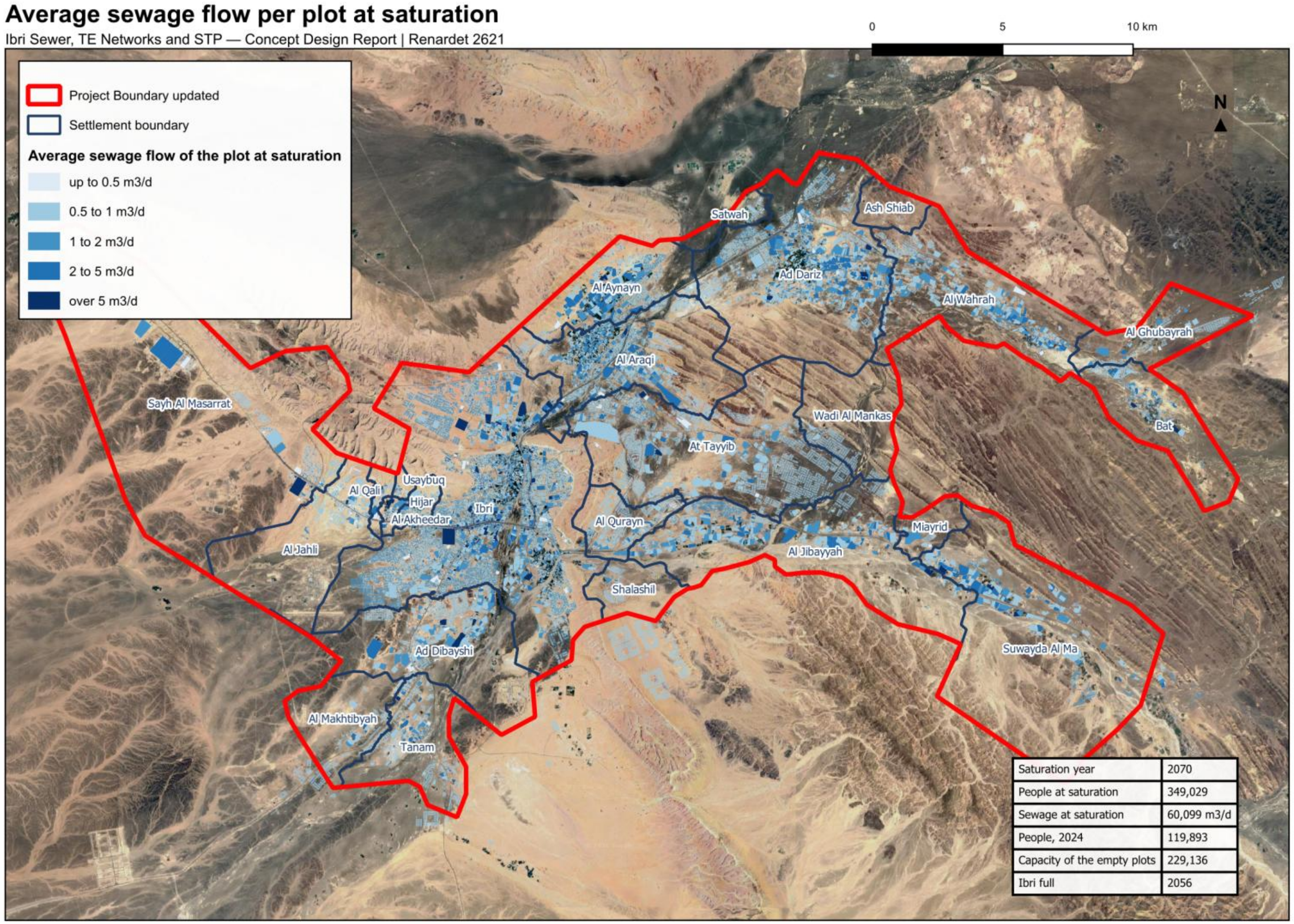


Figure 25 Average sewage flow of every plot at saturation. The network is sized on these flows, accumulated pipe by pipe.

16 Trade effluent, identified projects and special consumption

Discharges other than domestic sewage affect both the load on the treatment plant and the treatability of the influent. Where the ratio of chemical to biochemical oxygen demand exceeds the domestic range, a separate treatment line for high-strength wastewater is required if the volumes concerned are significant.

16.1 The large-consumer accounts

The electricity dataset carries 499 accounts on the Cost Reflective Tariff, which applies above a consumption threshold and says nothing about use: a shopping centre, a hospital, a mosque and a factory all fall in it. Each account has been placed by public records: named features in OpenStreetMap, reverse geocoding, the aerial imagery and the press. 205 are commercial, 135 governmental, educational, medical or religious, 15 industrial, 28 agricultural pumps and 6 telecommunications or utility; 110 stand more than eighty metres from any recorded feature and are carried as commercial. Each account carries the evidence for its placing and a confidence in the register delivered with this report.[1]

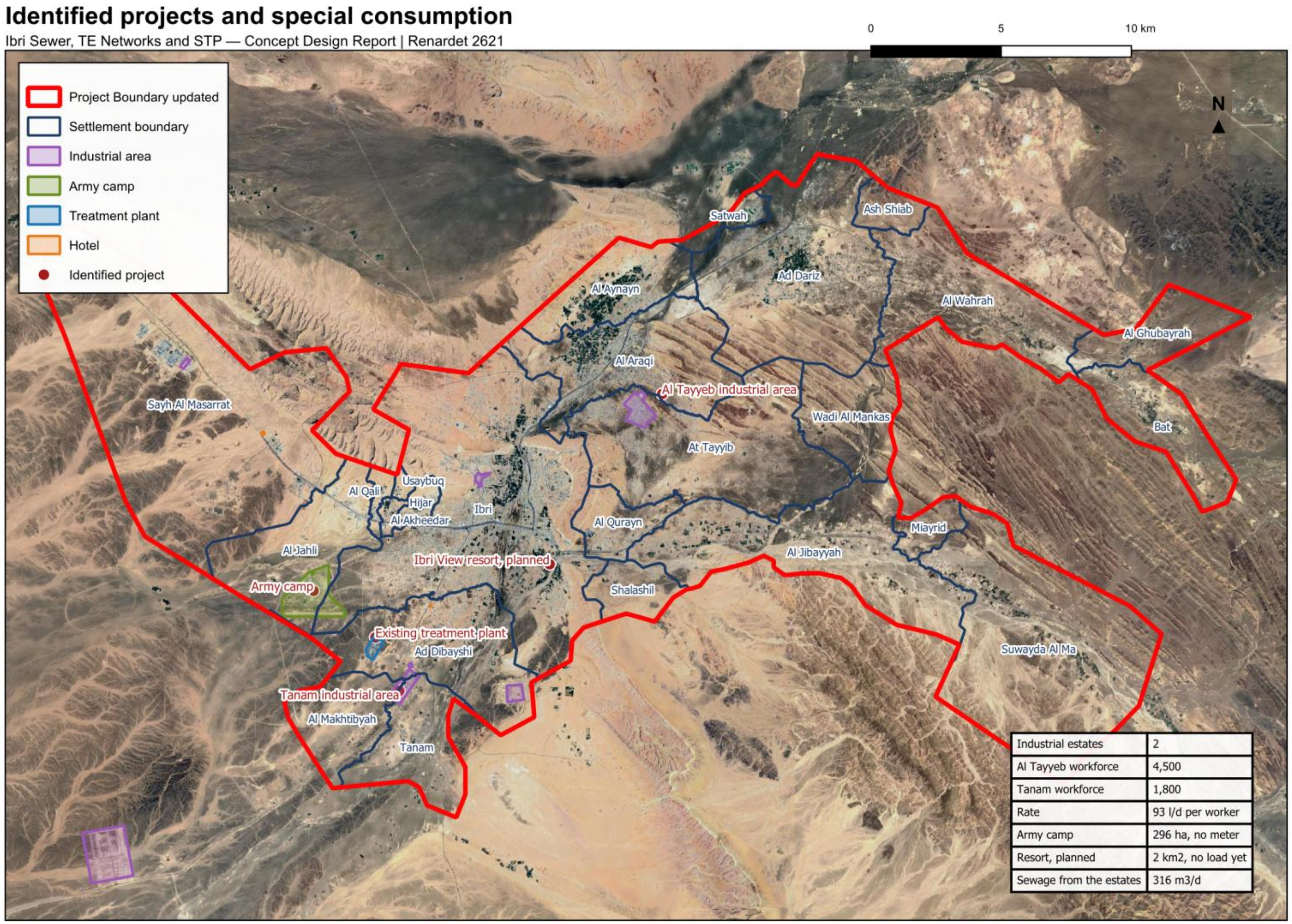
16.2 Identified projects

Beyond the accounts, the public record was searched for the projects the guideline keeps outside the population ratios: economic zones, camps and high-water users. Those found are listed below with the way each is treated.

Table 22 Identified projects and special consumption

Site	What it is	Treatment in the design
Al Tayyeb industrial area	94 ha, 202 industrial plots, 1,035 meters, in the north-east of the town	Special consumption, 4,500 workers at 93 l/d
Tanam industrial area	61 ha, 105 plots, 409 meters, beside the treatment plant	Special consumption, 1,800 workers at 93 l/d
Army camp, Northern Frontier Regiment	296 ha, west of the town, no meter in the dataset	Recorded; not connected to the network. Its occupancy and drainage are to be confirmed with the Ministry of Defence
Ibri View resort	A tourism development of 2 square kilometres at As Sulayf, announced, not yet in the cadastre	Recorded; no load until its plan is issued
Ibri Industrial City (Madayn)	10 square kilometres, 8.6 km south-west of the boundary, first phase opened 2024 on a collection tank	Outside the network; a source of tankered sewage to be confirmed from delivery records
Power station and solar plants	7.5 to 11 km north-west of the boundary, with a construction camp	Outside the network; a source of tankered sewage to be confirmed
Central slaughterhouse	Announced; site not yet fixed	High-strength discharge requiring pre-treatment; to be included when sited

[1] The largest groups are the Bawadi shopping centre with twenty accounts, the commercial street of Al Murtafa with nineteen, the police headquarters, the college campus and the hospital.



Industrial estates	2
Al Tayyeb workforce	4,500
Tanam workforce	1,800
Rate	93 l/d per worker
Army camp	296 ha, no meter
Resort, planned	2 km2, no load yet
Sewage from the estates	316 m3/d

Figure 26 The identified projects and special consumers: the industrial areas, the army camp, the treatment plant and the hotel sites recorded in OpenStreetMap, the two estates among them.

Sewage delivered by tanker is materially stronger than sewage arriving through the network. It is to be provided for separately in the plant load once the delivery records are received.[1]

[1] PAM-GUD-203, Table 31, page 68, gives tankered sewage at 350 to 1,050 mg/l BOD against 350 to 400 mg/l for network sewage.

17 Treated effluent demand and customers

Treated effluent is produced at 95 per cent of the plant inflow. A further ten per cent is lost within the distribution network.

$$Q_{\text{delivered}} = 0.90 \times 0.95 \times Q_{\text{inflow}} \tag{13}$$

Symbol	Meaning	Unit
Q delivered	treated effluent available to customers	m^3/d
Q inflow	flow entering the treatment plant	m^3/d

Customers are classified as public or private. Public consumers comprise the landscaping of highways, secondary roads, interchanges and roundabouts, and public parks. Private consumers comprise community parks, golf courses, private gardens, and nurseries and farms.

Table 23 Treated effluent demand for concept planning

Planting	Summer demand	Planting	Summer demand
Shrubs	20 to 40 l per plant per day	Ground cover	10 $l/m^2/d$
Palm trees	120 to 165 l per plant per day	Seasonal flowers	10 $l/m^2/d$
Other trees	40 to 80 l per plant per day	Grass	12 $l/m^2/d$
Hedges	10 l per metre per day	Roads and junctions	10 $l/m^2/d$

Demand varies seasonally, at 100 per cent from June to August, 75 per cent in spring and autumn and 50 per cent from December to February. The system is sized for the summer peak. Consumers whose average demand exceeds 500 cubic metres per day are assessed individually.[1]

The identification of customers, their present and potential demand and their development plans is in progress. The register will be presented in the next revision.

[1] PAM-GUD-201, Tables 21 to 23, pages 75 and 76. The rate for roads and junctions applies in the absence of specific vegetation information and is subject to municipality approval.

PART E

The existing system

18 Hydraulic assessment of the existing networks

18.1 Purpose

The existing sewer, force main and treated effluent networks are assessed against the flows arising across the design period, to establish which assets have capacity for the future flow, which require upgrading, and where the new works connect to them.

18.2 Approach

The assessment is carried out by hydraulic model. The existing network is verified before it is modelled: the datasets supplied provide geometry but not levels, and the diameter is recorded on part of the network only. The survey now in progress establishes both, and the model will be built on the surveyed data.

Each asset is then classified as having capacity for the design flow, requiring upgrading, or requiring replacement. Where an asset is retained, the point and manner of connection to the new works is designed.

18.3 Model

The wastewater network is modelled in SewerGEMS and the treated effluent network in WaterGEMS. Models are submitted in native editable format with the calculations and assumptions, and are updated following construction.[1]

The model is run for the design year at peak flow, for the ultimate condition, and for the opening years at low flow. The last of these establishes the period during which the network requires assisted cleansing, described in Section 22.4.

The assessment will be presented in the next revision, following completion of the survey.

19 Rehabilitation and upgrading

Rehabilitation of the existing systems forms part of the concept scope. The extent of it follows from the assessment in Section 18 and from the condition established by survey and closed-circuit television inspection.

Rehabilitation work received to date comprises contractor submissions for completed work orders at Al Sad, Khadil, Yanqul, Dhank and Hay Al Aqabah. These have been reviewed and recorded; they describe work already carried out rather than work required.

The rehabilitation schedule, with quantities and cost, will be presented in the next revision.

20 Verification of the Regional Master Plan

The Terms of Reference require the Regional Master Plan to be verified. The verification compares the flows established in Part D against those on which the master plan was based, and identifies and explains any difference.

The asset data supplied includes a treatment plant record shown as a design case with a capacity of 29,038 cubic metres per day, annotated as arising from the Regional Master Plan concept design and not yet approved. That figure is taken as the master plan position for comparison.[2]

[1] The software to be used is the subject of Section 5.4.

[2] STP_PT_IBRI.shp, record identifier 10, status Design, source recorded as Asset Planning.

The comparison depends on the design horizon, which is the subject of Section 5.1, and on the flow series described in Section 15.7. It will be presented in the next revision.

The data required by Nama Water Services Asset Management Planning for the updating of the master plan will be provided in the format the client specifies.

PART F

Options

21 Options methodology

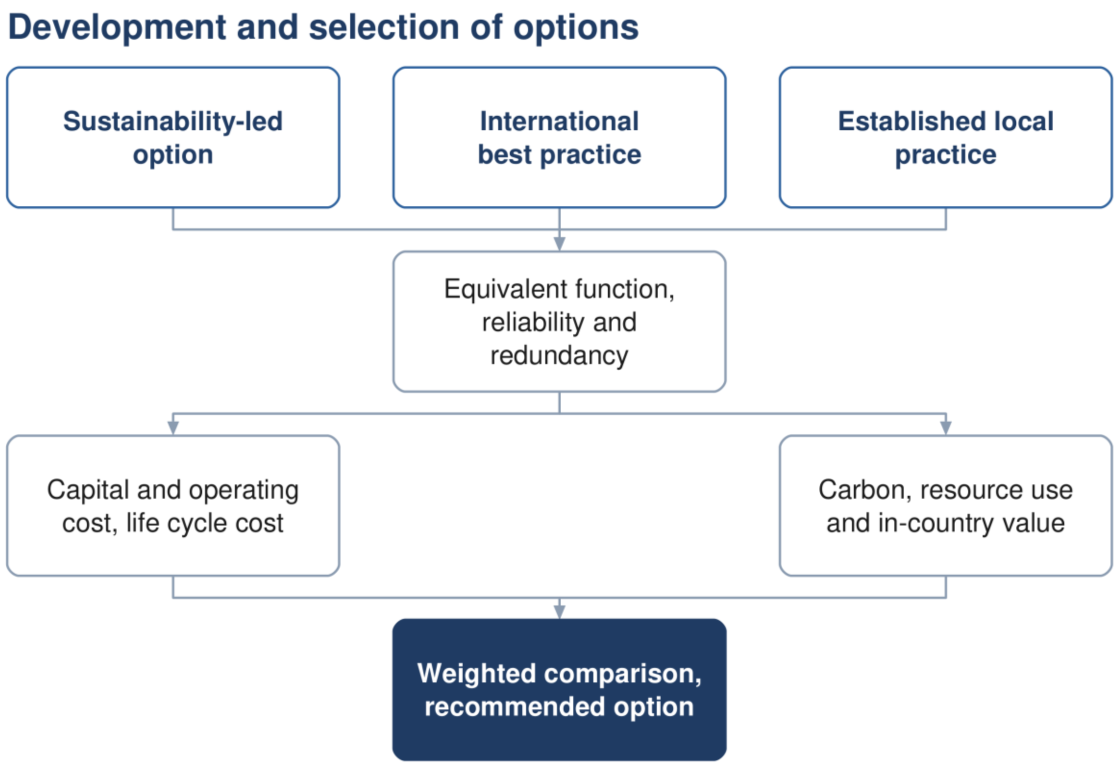


Figure 27 Development of the options and the basis on which they are compared.

21.1 Number and character of the options

Not fewer than three options are developed for each of the sewer network, the treated effluent network and the treatment plant. The guidelines indicate the character the options should take: one advancing environmental sustainability including nature-based solutions, one representing international best practice, and one based on established practice and technology available in the Sultanate.[1]

Each option is developed to equivalent functional requirements with comparable reliability and redundancy, so that the comparison between them is not influenced by differences in scope.

21.2 The approach proposed for each option

The guidelines describe the character of the three options but do not prescribe what distinguishes them in design terms. The following approach is proposed, and is offered for comment before the options are developed.

[1] PAM-GUD-201, Section 12.1, page 95.

Table 24 Proposed approach for each option

	Sustainability-led	International best practice	Established local practice
Network	Gravity maximised, accepting deeper excavation in order to avoid pumping	Layout optimised by model, with pumping where it reduces whole life cost	Conventional layout, with pumping where the topography requires
Treatment process	Lower-energy process, with nature-based polishing where the scale permits	Highest-performing process, smallest footprint	Process already established in Oman, simple to operate and maintain
Energy	Generation on site, with a self-sufficiency target	High-efficiency plant and advanced process control	Grid supply with standby generation
Reuse	Reuse maximised, discharge to wadi minimised	High reuse at high effluent quality	Reuse to the demand identified, surplus discharged
Materials	Low-carbon and locally sourced where performance permits	Specified for performance and durability	Standard specification
Consequence	Higher capital cost, lowest carbon and operating cost	Highest capital cost, least land, best effluent quality	Lowest capital cost, most land, highest operating cost

Two constraints apply across all three. Nature-based treatment is limited by the guidelines to small works, so it is available for outlying settlements or for polishing rather than for the main plant. And every option is developed to the same functional requirement and the same effluent standard, so that the difference between them lies in how the result is achieved and not in what is achieved.[1]

21.3 Basis of comparison

The options are compared on capital and operating cost, life cycle cost, carbon footprint over the project lifetime, resource efficiency, in-country value and the degree to which they employ nature-based solutions. The evaluation period is twenty-five years. The weighting applied to each parameter is set by Nama Water Services. The figure on the following page shows how each option is costed and compared.

[1] PAM-GUD-203, Section 10.5, page 101, limits constructed wetlands to works of approximately 500 cubic metres per day.

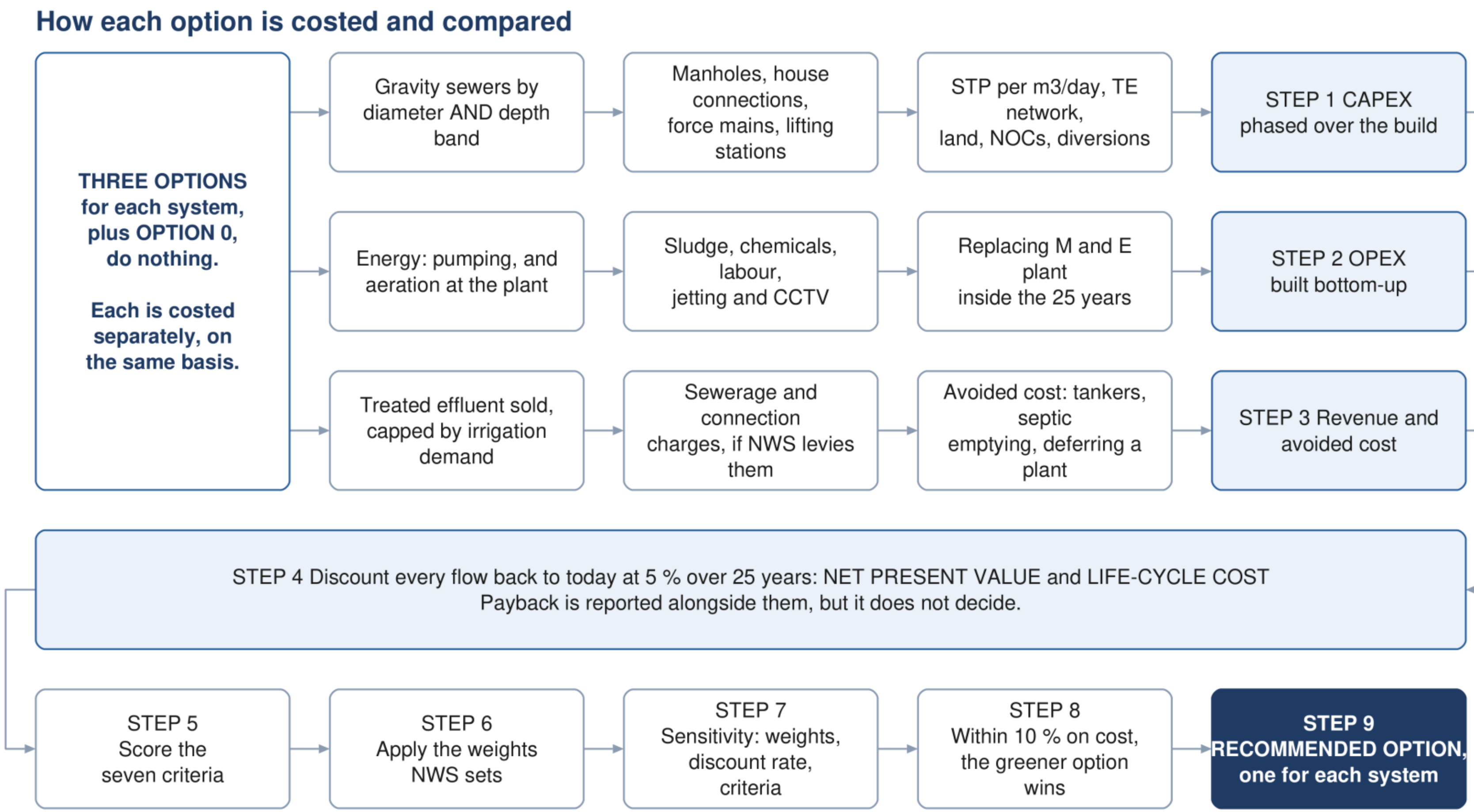


Figure 28 How each option is costed and compared. The three cost streams are established separately for every option, discounted to present value together, and only then scored against the seven criteria.

21.4 Selection

A weighted multi-criteria analysis compares the options against total lifetime cost, sustainability, social development and in-country value, adaptability and resilience, operability, constructability and environmental impact. Sensitivity is tested by varying the weighting between categories, the discount rate and the input design criteria. Where options fall within ten per cent of one another on total lifetime cost, the more sustainable option is adopted.[1]

22 Sewer network options

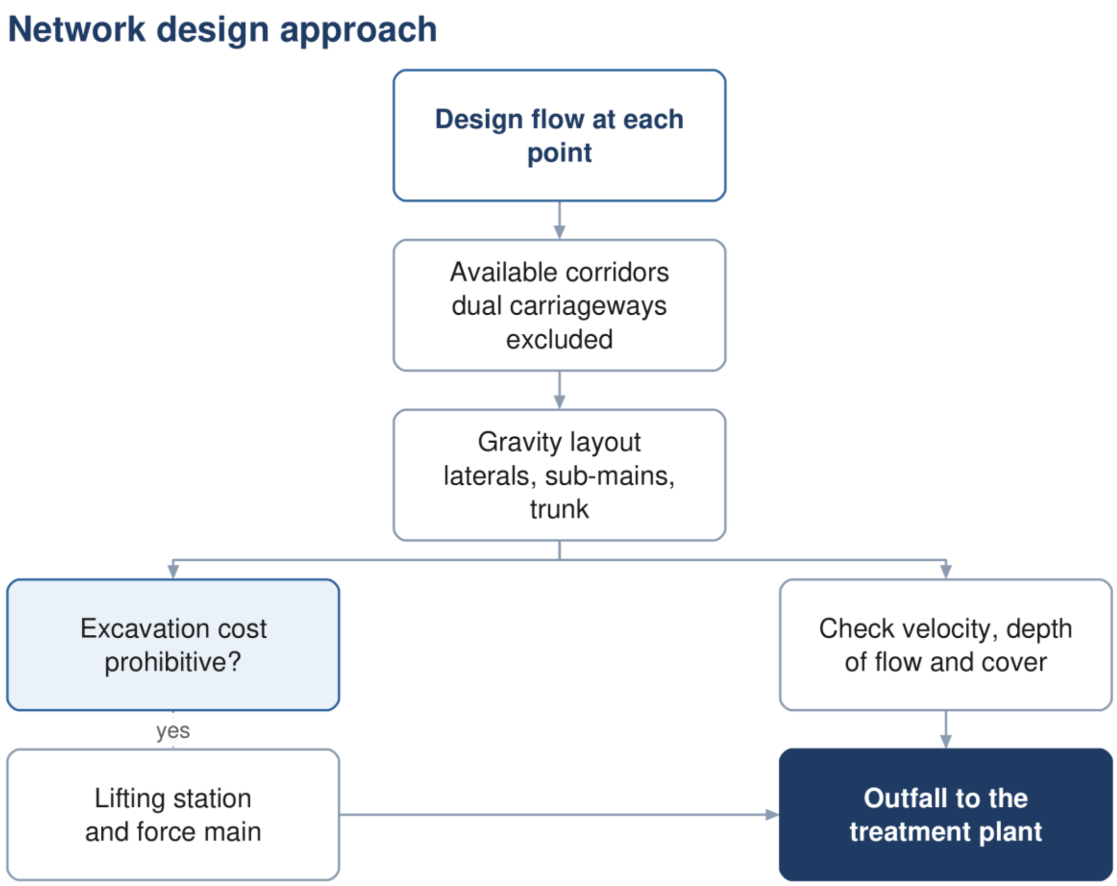


Figure 29 The network design approach, and the point at which a lifting station becomes necessary.

22.1 Principles

The network is laid out to convey the flow by gravity wherever that is feasible and cost effective, and to keep pumping to the minimum that the topography requires. The layout follows the road corridors, and the flow is directed towards the existing treatment plant site, which lies at a low elevation relative to the developed areas.

[1] PAM-GUD-201, Sections 12.6 to 12.9, pages 104 to 106.

22.2 Corridor constraints

Dual carriageways are excluded as sewer corridors, as they cannot be taken out of service for construction or maintenance. Crossings of a dual carriageway are made perpendicular and, where available, through an existing underpass. Wadi crossings are designed to the cover required by the guideline.

22.3 Network hierarchy

The network is arranged in three tiers, following the arrangement of the existing network in Ibri: laterals collecting from properties, sub-mains collecting from laterals, and a trunk main conveying the collected flow to the plant. Arranging the network this way limits the number of connections made directly to the trunk main and keeps each tier to a manageable size.

22.4 Early-year operation

In the years following commissioning the connected population is a fraction of the design population, and the flow in the network is correspondingly lower. Velocities in that period may fall below the self-cleansing value, and the network requires assisted cleansing until the flow is sufficient to scour the pipes.

The period during which this applies is established for each pipe from the flow series and the connection ratio, and is presented as a schedule of the pipes affected and the years concerned, so that the operating requirement is known before the network is handed over.

The network options and the cleansing schedule will be presented in the next revision.

23 Pumping and lifting stations

A lifting station is provided where the cost of excavation to maintain gravity flow becomes prohibitive. Each station lifts the flow to a level from which gravity conveyance resumes, discharging through a force main to a receiving manhole or to the treatment plant.

Table 25 Force main design criteria

Criterion	Value
Minimum velocity, raw sewage	0.75 m/s at minimum flow
Minimum velocity, intermittent flow	1.0 m/s
Minimum velocity, vertical mains	1.2 m/s
Maximum velocity	2.5 m/s
Minimum internal diameter	75 mm
Gradient, rising	1 in 500
Gradient, falling	1 in 300
Retention time	as short as the alignment permits

Retention in a force main allows sulphide to form, which is both a corrosion and an odour risk in this climate. Alignments are therefore kept short, discharges are submerged to limit turbulence, and the need for dosing is assessed for each station.[1]

Surge analysis is carried out for each force main in approved software, and the protection required is established from it.

[1] PAM-GUD-203, Sections 8.2.1 and 11.5.3, pages 50 and 181.

24 Treated effluent network options

The treated effluent network conveys the product of the treatment plant to the customers identified in Section 17. The network is sized for the summer peak demand, with storage of not less than twenty-four hours.

Where demand is below the volume produced, provision is made for the disposal of the excess, described in Section 26.

The network options will be presented in the next revision, following confirmation of the customers and their demand.

25 Treatment plant options

25.1 Capacity and phasing

The plant is sized on the flow established in Part D with the ten per cent design margin applied, and is built in phases so that capacity follows demand. The capacity of the first phase, and the years at which subsequent phases are required, follow from the flow series and are presented with it.

25.2 Process selection

Process selection is governed by the effluent standard. The total nitrogen limit described in Section 13 requires full nitrification and denitrification, which narrows the technologies that can be considered. Land area, energy consumption, operating complexity and whole life cost distinguish the remaining options.

Table 26 Indicative land requirement by process

Process	Area, m^2 per m^3/d
Membrane bioreactor	0.45 to 0.9
Sequencing batch reactor	0.9 to 1.8
Moving bed biofilm reactor	0.9 to 1.8
Integrated fixed film activated sludge	1.2 to 2.5
Conventional activated sludge and extended aeration	1.8 to 3.6

The areas are indicative and are used for planning. The land required includes the process units, the sludge facilities, the buffer zone and provision for the energy-efficiency measures described in Section 34.[1]

25.3 Siting

The site is assessed against the criteria in the wastewater guideline, covering access, physical characteristics, environmental and climatic impact, social impact and cost. Two criteria carry stated requirements: the site is assessed against the twenty-five and one hundred year flood levels, and the plant is to remain operational during floods.

The buffer distance to residential areas for a plant of this size is established from odour dispersion modelling rather than from a fixed figure, and lies between 300 and 1,000 metres measured to the five odour unit contour. The modelling is described in Section 30 and precedes the confirmation of the site.[2]

The plant options, the site assessment and the phasing will be presented in the next revision.

26 Excess effluent, emergency provisions and tankers

26.1 Tanker reception

A proportion of the wastewater arising in the area is collected by tanker and delivered to the treatment plant. Sewage delivered this way is stronger than sewage arriving through the network, and its arrival is concentrated in the working day. A dedicated reception facility is provided, with screening, grease removal, sampling before acceptance and flow equalisation.

[1] PAM-GUD-203, Table 28, page 64.

[2] PAM-GUD-201, Table 8, pages 43 and 44.

26.2 Excess treated effluent

Where the effluent produced exceeds the demand, provision is made for its disposal. Discharge to a wadi requires the effluent to meet Class A of Ministerial Decision 145/1993, and where the wadi discharges to the sea the limits for ammonia, nitrogen and phosphorus of Ministerial Decision 159/2005 apply in addition. Any such discharge is subject to the approval of the Environment Authority and the Authority for Public Services Regulation.[1]

26.3 Emergency provisions

Provision is made for the diversion of raw sewage in the event that the plant is out of operation, and for emergency storage. Every pumping station is provided with an emergency overflow to prevent flooding of the station or of connected properties; overflows are subject to the approval of the Environment Authority.

27 Sludge management strategy

The Oman Sludge Management Plan, as reproduced in the wastewater design guideline, identifies a sludge treatment centre performing composting at Ibri as the solution for the Governorate of Adh Dhahirah. The plant is therefore to be designed to receive and process sludge arising beyond its own catchment, and the land area, buffer distance and vehicle access are established accordingly.[2]

Sludge is to be reused unless no form of reuse is possible, and its quality is to comply with the heavy metal limits of Ministerial Decision 145/1993. Where disposal to landfill is required, the receiving authority sets acceptance criteria including a minimum solids content of eighty per cent. That content is above what mechanical dewatering alone achieves, and drying or composting is therefore required.

Dewatering facilities are provided on site. The strategy, with the quantities arising and the disposal route for each, will be presented in the next revision.

[1] PAM-GUD-203, Section 10.2.4.3, pages 72 and 73. The substituted limits are materially tighter than Class A for phosphorus and ammoniacal nitrogen.

[2] PAM-GUD-203, Table 67, page 136.

PART G

Assessment and appraisal

28 Ground conditions

Geophysical investigation is required for all infrastructure and geotechnical investigation for all above-ground assets. At this stage the purpose is to establish a site-wide ground model and a range of parameters sufficient to select foundation types and platform levels, and to identify the risks that the detailed investigation must resolve.

Boreholes at the treatment plant and lifting station sites are taken to a minimum depth of fifteen metres or to a competent stratum, and are closely spaced at wet wells, valve chambers and electrical rooms. Groundwater monitoring is required, as the depth to groundwater governs both the excavation method and the infiltration allowance.

The ground model and the concept geotechnical report will be presented in the next revision.

29 Flood protection

The study area is crossed by a wadi system, and flood behaviour governs both the alignment of the works and the siting of the treatment plant. The site is assessed against the twenty-five and one hundred year flood levels, and the plant is to remain operational during floods.

Pumping station floor levels, transformers and standby generators are set not less than 300 millimetres above the one in fifty year flood level. Wadi crossings are designed with a minimum cover of 1.5 metres to the crown of the pipe.

The flood protection assessment will be presented in the next revision.

30 Odour assessment

Odour governs the buffer distance between the treatment plant and residential development, and for a plant of this size the buffer is established from dispersion modelling rather than from a fixed distance. The modelling establishes the distance to the five odour unit contour, using site meteorological records and the treatment processes proposed.[1]

Odour control is provided at the inlet works, the sludge facilities and the pumping stations, with the treatment train selected from the assessment. Compliance is verified by continuous monitoring at the site boundary.

The dispersion modelling and the resulting buffer will be presented in the next revision, and precede confirmation of the plant site.

31 Climate resilience

The works are assessed for resilience to the climate conditions anticipated over their service life. The assessment considers a fifty-year horizon at two and at four degrees Celsius of warming, and addresses the effect of changes in rainfall frequency and intensity on site selection and on the design of the infrastructure.[2]

The assessment will be presented in the next revision.

[1] PAM-GUD-201, Table 8, page 43; PAM-GUD-203, Table 90, page 170, which sets five odour units per cubic metre at the site boundary where the surrounding area is sensitive.

[2] PAM-GUD-201, Section 4.3, page 33.

32 Environmental and social assessment

An environmental impact assessment is required for the treatment plant and its associated works. The Environment Authority is informed of the project, its objectives and its anticipated impact, and is consulted on the location of the facilities.

Each option is assessed against resource use, ecosystem disruption, pollution risk, climate resilience, socio-economic impact and land use, and the assessment forms part of the comparison described in Section 38.

The scope required at this stage is the subject of Section 5.5.

33 Utility interfaces and approvals

33.1 Approach

The proposed alignments are superimposed on the service records of each authority holding assets in the area. Where a conflict cannot be avoided by re-routing within the corridor, the cost of relocating the proposed works is compared with the cost of relocating the existing service, and the agreement of the owning authority is obtained for whichever is adopted.

33.2 Records held and required

The potable water network is available from the dataset described in Section 7.2 and provides 647.8 kilometres of mains within the study area. Records for electricity distribution, telecommunications and, where present, gas and fuel pipelines are being requested from the respective owners.

The electricity data held records the position of consumer connections. It does not record the routes of the distribution cables, which are required for clash assessment and are requested separately.[1]

33.3 Clearances

Table 27 Separation from other services

Situation	Requirement
Force main to water main, horizontal	3.0 m
Force main crossing a water main	the force main passes beneath, with 450 mm vertical clearance
Shallow sewer beneath a major road	3.0 m horizontal clearance
Another service in the same trench	placed on a separate bench on undisturbed ground

Beyond these, the clearance applied is that specified by the authority owning the service.

33.4 Trial pits

Fifty trial pits are provided for. Their locations are selected at road intersections, along the routes of major existing services and along the expected routes of the trunk sewers and force mains. The programme is agreed with Nama Water Services and the municipal excavation approvals obtained before work begins.

[1] The account dataset described in Section 7.3 comprises point locations of metered connections.

33.5 Approvals register

Approvals and no objection certificates are required from the authorities listed below. A register is maintained recording the authority, the consent required, the date of application and the current position, and is reported to Nama Water Services.

Table 28 Authorities from which approvals are required

Authority
Ministry of Housing and Urban Planning
Environment Authority
Ministry of Agriculture, Fisheries and Water Resources
Directorate General of Roads
Regional electricity distribution company
Oman Telecommunications Company
Royal Oman Police
Ministry of Heritage and Tourism
Municipality of Ibri

34 Sustainability

The carbon footprint of each option is evaluated for both construction and operation in accordance with recognised greenhouse gas accounting standards, and is expressed in tonnes of carbon dioxide equivalent per year and per cubic metre of effluent produced.[1]

Because the greater part of operational emissions arises from electricity consumption, the measures that reduce carbon are largely the same as those that reduce operating cost: conveying the flow by gravity wherever possible, minimising lift, selecting efficient equipment, and generating renewable energy on site. Provision for photovoltaic generation within the plant perimeter is included in the land requirement.

Resource efficiency, in-country value and the use of nature-based solutions are assessed for each option and carried into the comparison in Section 38.

35 Cost

35.1 Basis

Item	Basis
Accuracy	plus or minus twenty per cent at concept stage
Measurement	CESMM3
Rates	recent tender and contract rates, and rates from Nama Water Services pre-investment appraisals, escalated to the base date of the estimate
Price basis	current prices at the date of the estimate
Presentation	by system element

35.2 Scope of the estimate

The estimate covers the works, the associated costs and the provisions, as set out below.

- **Collection network —** excavation by depth and by ground condition, trench support, bedding and backfill, pipework, manholes, property connections, road reinstatement, traffic management, crossings, and testing.
- **Lifting stations —** land, civil structures, pumps and station pipework, electrical supply and standby generation, control and telemetry, odour control, and the force main with its chambers and valves.
- **Treatment plant —** land, site preparation and flood protection, inlet works, tanker reception, biological treatment, clarification, tertiary treatment, disinfection, the sludge line, chemical systems, odour control, electrical and control installations, buildings, and commissioning.
- **Treated effluent network —** pipework, storage, boosting, filling stations and customer connections.
- **Associated costs —** design and supervision, survey and investigation, environmental assessment, consents and land, and physical and price contingency.

35.3 Operating cost

Operating cost is built from the duty of each asset rather than taken as a proportion of its capital cost. For a wastewater system the two largest items are energy and sludge, and neither follows capital value.

- **Energy —** pumping at each lifting station from its duty flow and head, and aeration at the treatment plant from the oxygen demand.

[1] PAM-GUD-201, Section 12.5.1, pages 98 to 100, which cites ISO 14064 and the Greenhouse Gas Protocol and gives a benchmark of 1.17×10^{-3} tonnes of carbon dioxide equivalent per cubic metre of treated effluent.

- **Sludge —** thickening, dewatering, transport and disposal, at the quantity the process produces.
- **Chemicals —** coagulant, polymer and disinfectant at the dose rate and the flow treated.
- **Labour —** the establishment required to operate and maintain the plant, the stations and the network.
- **Maintenance —** jetting and closed-circuit television survey of the network, desludging, odour and corrosion control, and planned replacement of mechanical and electrical plant falling within the evaluation period.

Mechanical and electrical plant has a shorter service life than the civil works, so its replacement falls inside the twenty-five year period and is carried as a cost in the year it occurs.[1]

35.4 Life cycle cost and net present value

Capital cost, operating cost, replacement cost and revenue arise in different years, so they are not comparable until they are brought to a common date. Each is discounted to present value at the rate the guideline sets, and the options are compared on the resulting totals.[2]

$$\mathrm{NPV} = \sum_{t=0}^{n} \frac{C_t}{(1+r)^t} \tag{14}$$

Symbol	Meaning	Unit
NPV	net present value of the option	OMR
C t	net cash flow in year t: revenue and avoided cost less capital, replacement and operating cost	OMR
r	discount rate, 0.05	—
t	year, counted from the base date	—
n	evaluation period, 25 years	—

Total life cycle cost is the present value of capital, replacement and operating cost over the same period, and is the quantity against which the ten per cent band in Section 21.4 is applied. Payback is reported alongside it as an indicator, but it does not determine the recommendation: for a wastewater scheme a large part of the benefit is cost avoided rather than revenue earned, and a payback period computed on revenue alone would misstate the case.[3]

The estimate, the discounted comparison and the payback figures will be presented with the options in the next revision.

36 Risk

A risk register is established at this stage and maintained through the project. Each risk carries a description, a likelihood, an impact expressed in cost or time, an owner and the mitigation adopted. Risks that fall on one option and not on another are identified as such, as they affect the comparison rather than only the total.

[1] PAM-GUD-201 sets a design life of 25 years for assets generally; rotating plant and instrumentation are replaced at least once within that period.

[2] PAM-GUD-201, page 57, states that the twenty-five year planning life is also the period over which net present value is calculated for the comparison of schemes; the discount rate of five per cent is given at pages 95 to 96.

[3] Net present value and life cycle cost are not defined by equation in the Nama Water Services guidelines. The formulation above follows ISO 15686-5, with the period and discount rate taken from PAM-GUD-201.

37 Value engineering

A formal value engineering study is required at the concept and preliminary stages for a treatment plant or pumping station above the stated threshold, carried out by an independent certified consultant. The study is arranged within the concept programme and its outcome reported.[1]

38 Comparison and recommendation

The options are compared by the method described in Section 21.3. The comparison presents, for each option, the capital cost by phase, the operating cost by year, the net present value over twenty-five years, the carbon footprint, and the assessment against each of the remaining criteria, together with the results of the sensitivity tests.

The net present value of each option is computed by the equation of Section 35.4, at five per cent over twenty-five years.

The comparison and the recommended option will be presented in the next revision.

[1] PAM-GUD-201, Table 27, page 93, and its accompanying note.

PART H

Delivery

39 Implementation roadmap

An implementation roadmap is prepared for the recommended option, defining the scope of the subsequent design stages, the procurement route for each element, the phasing of construction, and the framework by which performance is monitored once the works are in service.

The roadmap follows the selection of the recommended option and will be presented with it.

40 Contracting strategy

The contracting strategy establishes how the works are packaged and procured. It is developed with Nama Water Services in a dedicated workshop, and considers the division of the works into packages, the procurement route for each, and the interfaces between them.

41 Project integration

A project integration plan sets out how the water and wastewater components are coordinated through design, construction and commissioning. It also addresses the rehabilitation or replacement of the existing potable water network where its performance is found to be unsatisfactory.[1]

42 Conclusions

The design basis for the wastewater and treated effluent systems is established and is set out in Part C. The data supplied has been assessed, and the datasets that are usable, those that require correction and those that relate to areas outside the project have been identified in Part B.

The existing wastewater assets within the study area comprise 111.6 kilometres of constructed gravity sewer and 10.0 kilometres of constructed force main, together with a further 268.2 kilometres of gravity sewer, pumping main and treated effluent main recorded as proposed. The potable water network within the area comprises 647.8 kilometres of mains. The number of properties on each plot and the use of each plot have been established from 33,971 electricity meters and a satellite image, the occupancy rate has been derived for each settlement, and every plot carries a population and an average sewage flow for 2024 and for every year to the saturation of its settlement. The study area is saturated in 2070 at 349,029 people and 60,099 cubic metres of sewage a day.

A topographic and utility survey covering the whole study area is in progress. It will establish the levels, diameters and condition that the supplied datasets do not carry, and its completion is the principal step between this revision and the next.

42.1 Recommendations

It is recommended that Nama Water Services confirm the seven matters set out in Section 5 and the departures in Section 10.1, and that the datasets identified in Section 7 as requiring correction or clarification be resolved, so that the flow series, the options and the appraisal can be completed on an agreed basis.

[1] PAM-GUD-201, Section 13, page 107.

43 Appendices

Appendix	Content
A	Population, land use and flow: the working behind Part D. Issued with this revision
B	Design criteria, with references. Next revision
C	Drawings and figures. Next revision
D	Register of matters requiring confirmation. Next revision

The plot layer, the settlement layer, the meter layer, the register of large-consumer accounts and the growth workbook accompany this report in digital form.

Appendix A

Population, land use and flow: the working

A1 Electricity meters: tariff to category

Each tariff in the electricity dataset is assigned to one of the guideline's categories. The large-consumer tariff is resolved account by account in A2.

Table 29 Tariffs and the category adopted

Tariff	Meters	Category	People
Primary Account Tariff	10,973	Domestic	one property
Primary Account Tariff (with National Subsidy)	5,272	Domestic	one property
Additional Account Tariff	6,344	Domestic	one property
Commercial	9,385	Non-domestic	none
Fisheries	6	Non-domestic	none
Tourism	1	Non-domestic	none
Government	966	Governmental	none
MOD	1	Governmental	none
Agricultural	523	Agricultural	none; no sewage
Industrial	1	Special	none
CRT Seasonal	298	Large consumer, see A2	none
CRT Time of Use	192	Large consumer, see A2	none
CRT Fixed Rate	9	Large consumer, see A2	none

Of the 33,971 meters, 30,931 fall inside a plot, 2,914 were assigned to the nearest plot within fifteen metres, and 126 lie further from any plot; they are listed with their settlement and carry no load in the plot table.

A2 The large-consumer accounts

The 499 accounts on the Cost Reflective Tariff were placed by clustering them at 120 metres, matching each cluster and each single account against the named features of OpenStreetMap within eighty metres, reverse geocoding the rest, and inspecting the largest clusters on the aerial imagery. The register delivered with this report carries, for each account, the use found, the category adopted, the evidence and a confidence.

Table 30 Large-consumer accounts by the use found

Use found	Accounts	Category adopted	Typical sites
Commercial	205	Non-domestic	Bawadi shopping centre, the Al Murtafa souq, the Araqi strip, hypermarkets, banks, fuel stations
Government	67	Governmental	Police headquarters, post office, directorates
Education	37	Governmental	The university campus, the college, schools
Health	15	Governmental	Ibri Hospital, clinics
Religious	16	Governmental	Mosques
Industrial	15	Special	The Tanam estate
Agricultural	28	Agricultural, no load	Farm pumps at Aynayn
Telecommunications, utility	6	Non-domestic	Exchanges, water works, masts
Unresolved	110	Non-domestic	No recorded feature within eighty metres

Confidence: 119 accounts certain, 194 likely, 186 placed on weaker evidence. The categories adopted affect only where the non-domestic and governmental water is placed, not its total.[1]

[1] Certain: a named feature within twenty metres, or inspected. Likely: a named feature within eighty metres, or a nearest feature by reverse geocoding. Weaker: the surroundings only.

A3 Occupancy, properties per plot and home share by settlement

Table 31 The settlement rates used in Part D: derived from the data, and adopted

Settlement	Domestic properties, 2024	Pop. 2024	Empty plots	of which counted as future homes	Occupancy derived	Occupancy adopted	Properties per home plot derived	adopted	Home share derived	adopted	Capacity, people
Ibri	11,052	67,106	14,203	10,469	6.07	6.07	1.33	1.33	0.87	0.87	73,299
Ad Dariz	2,439	11,850	4,888	3,219	4.86	4.86	1.20	1.20	0.90	0.90	16,854
Al Araqi	2,466	10,696	3,360	2,463	4.34	4.34	1.32	1.32	0.84	0.84	11,832
Al Aynayn	889	4,554	1,780	1,362	5.12	5.12	1.14	1.14	0.94	0.94	7,459
Al Wahrah	716	3,351	2,000	1,373	4.68	4.68	1.22	1.22	0.91	0.91	7,109
At Tayyib	833	3,348	10,012	8,640	4.02	4.02	1.09	1.09	0.92	0.92	34,675
Al Jibayyah	524	2,686	2,784	1,966	5.13	5.13	1.26	1.26	0.89	0.89	11,306
Bat	418	2,557	879	355	6.12	6.12	1.17	1.17	0.88	0.88	2,226
Ad Dibayshi	681	2,411	2,430	1,887	3.54	4.00	1.16	1.16	0.90	0.90	7,920
Tanam	403	2,116	2,045	1,556	5.25	5.25	1.24	1.24	0.67	0.67	6,779
Suwayda al Ma	301	1,559	1,593	937	5.18	5.18	1.60	1.60	0.82	0.82	6,369
Hijar	352	1,033	375	211	2.93	4.00	1.69	1.69	0.90	0.90	1,288
Al Ghubayrah	161	631	1,448	1,094	3.92	4.00	1.12	1.00	0.84	0.90	3,940
Al Qurayn	147	549	2,051	1,830	3.73	4.00	1.14	1.00	0.93	0.90	6,588
Sayh al Masarrat	457	466	1,745	1,384	1.02	4.00	2.52	1.00	0.92	0.90	4,984
Al Jahli	344	415	1,235	1,023	1.21	4.00	2.16	1.00	0.78	0.90	3,684
Satwah	27	263	181	79	9.73	4.00	1.23	1.00	0.87	0.90	284
Al Akheedar	120	198	120	45	1.65	4.00	2.12	1.00	0.53	0.90	160
Al Qali	56	191	220	157	3.41	4.00	2.20	1.00	0.92	0.90	564
Shalashil	38	130	3,686	3,250	3.42	4.00	1.11	1.00	0.91	0.90	11,700
Al Makhtibyah	91	122	1,523	1,192	1.34	4.00	1.15	1.00	0.90	0.90	4,292
Usaybuq	21	91	60	39	4.33	4.00	1.25	1.00	0.87	0.90	140
Wadi al Mankas	8	52	1,580	1,394	6.49	4.00	1.30	1.00	0.87	0.90	5,020
Ash Shiab	11	47	65	13	4.27	4.00	1.10	1.00	0.87	0.90	48
Miayrid	4	34	246	171	8.49	4.00	1.30	1.00	0.87	0.90	616

The occupancy adopted is the rate derived, held to a floor of 4.0 and a cap equal to the highest rate among the settlements of two thousand or more people. A settlement of fewer than a thousand people in 2024 takes an occupancy of 4.0, one property per plot and a home share of 0.9, whatever its meters return: it has no sample to measure on, and it does not attract second dwellings.[1]

A4 The plot and settlement layers delivered with this report

The plot layer carries, for each of the 77,265 plots, the fields below. It is the load table of the design: the network model reads the flow of each plot from it, and every figure in Part D is a sum over it. The settlement layer carries the same at settlement level, with the rates and the five-year steps.

Table 32 Fields of the plot layer

Field	Content
Name	cadastral identifier, as received
Moh_Classi, Classes	class code and class name of the cadastre, as received
Buiding_St	built or future, as received
SETTLE	the settlement the plot belongs to
AREA_M2	plot area, m²
N_ACC	meters on the plot, all tariffs
N_DOM, N_DOMADD	meters on the primary and subsidised tariffs; on the additional-account tariff

[1] Properties per home plot is measured over built plots whose meters are more than two thirds domestic and fewer than fifteen. Home share is the proportion of built, metered plots of home shape that are homes. Empty plots counted as future homes are those of 200 to 1,000 square metres, compact, not a strip, and not a grove, an industrial plot or heritage.

Field	Content
N_COM, N_GOV, N_AGR, N_CRT, N_IND	meters on the commercial tariff (with fisheries and tourism), the government tariff (with defence), and the agricultural, large-consumer and industrial tariffs
G_DOM	domestic meters, one property each (the primary, subsidised and additional tariffs)
G_NDOM, G_GOV, G_SPEC, G_AGR	non-domestic, governmental, special and agricultural meters, the large consumers placed by their identified use
DERIVED	use of the plot (Section 14.5)
WHYC	the rule that set the use
ESTATE	the industrial estate the plot lies in, if any
WORKERS	workers assigned to an estate plot, by area
HOMESHAPE	1 if the plot is home-shaped: 200 to 1,000 m², compact, not a strip
COMPACT, ASPECT	area against the smallest enclosing rectangle; that rectangle's aspect ratio
NDVI_MEAN, NDVI_SHARE, GREEN_M2	the satellite vegetation test: mean index, share of green pixels, green area in m²
OR_S	occupancy adopted for the settlement, persons per property
PPP_S	properties per home plot adopted for the settlement
HOMESH_S	home share adopted for the settlement
U_NDOM, U_GOV	non-domestic water per shop meter and governmental water per government meter in the settlement, l/d
POP	people in 2024: G_DOM × OR_S
W_DOM, W_NDOM, W_GOV, W_SPEC, W_TOT	water by stream and in total, m³/d, 2024
S_DOM, S_NDOM, S_GOV, S_SPEC	sewage by stream, m³/d: 0.85 of the domestic water, 0.54 of the rest
QADF	average dry-weather sewage flow of the plot in 2024, m³/d: the sum of the four streams
FUT_CAP	1 if an empty plot counts as a future home
FUT_PROPS	properties expected per counted plot: the settlement's ratio times its home share
SPREAD_W	weight for placing the settlement's growth: the plot's area capped at 1,000 m² on an empty plot of up to 2,000 m² that is not a grove, industrial or heritage; zero on every other plot; its share is this weight over the settlement's sum
POP_2030, POP_2055, POP_ULT	people on the plot in 2030, 2055 and the saturation year
Q_2030, Q_2055, Q_ULT	average sewage flow in 2030, 2055 and the saturation year, m³/d; a future plot at 171.3 l/d per person
SAT_YEAR	the year the plot's settlement is full
ULT_YEAR	the saturation year of the study area

Table 33 Fields of the settlement layer

Field	Content
SETTLE, AREA_KM2	name; area in km²
WB_2024, WB_2100	population of the Inception Report series in 2024 and 2100
PROPS	domestic properties (meters) in 2024
OR_RAW, OR_S	occupancy derived and adopted
PPP_RAW, PPP_S	properties per home plot derived and adopted
HOMESH_RAW, HOMESH_S	home share derived and adopted
SMALL	1 if under 1,000 people in 2024: the rule of Section 14.4 applies
BUILT_PL, EMPTY_PL, HOME_MEAS	built plots; empty plots; home plots the ratio was measured on
CAP_PLOTS, CAP_POP	empty plots counted as future homes; the people they hold at saturation
G_NDOM, NDOM_POOL, G_GOV, GOV_POOL, G_AGR, G_SPEC	meters by category; of the shop and government meters, those outside the estates that share the pool
WORKERS	workers of the industrial estates in the settlement
U_NDOM, U_GOV	non-domestic water per shop meter, governmental water per government meter, l/d
POP_2024	people in 2024
W_DOM, W_NDOM, W_GOV, W_SPEC, W_TOT	water by stream and in total, m³/d, 2024
S_DOM, S_NDOM, S_GOV, S_SPEC, Q_2024	sewage by stream and in total, m³/d, 2024

Field	Content
POP_2025 … POP_2070, Q_2025 … Q_2070	people and average sewage flow at five-year steps to the saturation year
POP_ULT, Q_ULT	people and flow in the saturation year of the study area
SAT_YEAR, SAT_POP, SAT_Q	the year the settlement is full, and its people and flow then
SAT_OWN	the year it would fill on its own growth alone; 0 if not before 2100
IN_PEOPLE, OUT_PEOPLE, MAIN_TO, INSHARE	people received from and sent to other settlements by the saturation year; the main receiver; the share of capacity taken by overflow
ULT_YEAR, UNHOUSED	the saturation year of the study area; growth of the series to 2100 with nowhere to go

A5 The overflow routes in full

Every route carrying fifty people or more at saturation. The donor's year is the year its own empty plots are full; the receiver starts in the first year it takes in overflow from any settlement, which can be before this donor is full, and is itself full in the last column.

Table 34 Overflow routes

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Ibri	At Tayyib	22,078	2056	2060	2068
Ibri	Shalashil	11,471	2056	2056	2061
Ibri	Al Qurayn	5,856	2056	2056	2060
Ad Dariz	Wadi al Mankas	4,965	2062	2062	2068
Al Araqi	At Tayyib	4,873	2057	2060	2068
Ibri	Ad Dibayshi	4,078	2056	2062	2063
Ibri	Al Jibayyah	3,927	2056	2068	2070
Bat	Al Ghubayrah	2,493	2052	2052	2070
Ibri	Al Makhtibyah	2,441	2056	2063	2069
Ibri	Tanam	2,010	2056	2063	2069
Al Araqi	Al Aynayn	1,790	2057	2057	2060
Al Aynayn	At Tayyib	1,639	2060	2060	2068
Al Araqi	Al Jibayyah	1,410	2057	2068	2070
Ibri	Sayh al Masarrat	1,225	2056	2063	2069
Ad Dibayshi	Tanam	1,004	2063	2063	2069
Ibri	Al Jahli	915	2056	2054	2063
Ad Dariz	Al Wahrah	863	2062	2062	2070
Al Araqi	Suwayda al Ma	730	2057	2070	2070
Al Aynayn	Al Jibayyah	600	2060	2068	2070
Ad Dariz	Suwayda al Ma	551	2062	2070	2070
Hijar	Al Makhtibyah	519	2053	2063	2069
Hijar	Al Jahli	518	2053	2054	2063
Al Jahli	Al Makhtibyah	507	2063	2063	2069
Ibri	Suwayda al Ma	490	2056	2070	2070
At Tayyib	Miayrid	441	2068	2068	2070
Ibri	Ad Dariz	440	2056	2057	2062
Al Aynayn	Ad Dariz	347	2060	2057	2062
Al Aynayn	Suwayda al Ma	311	2060	2070	2070
Al Akheedar	Al Qali	297	2036	2036	2054
Ad Dariz	Al Ghubayrah	258	2062	2052	2070
Al Qurayn	At Tayyib	244	2060	2060	2068
At Tayyib	Suwayda al Ma	229	2068	2070	2070
Al Wahrah	Suwayda al Ma	229	2070	2070	2070
Ad Dibayshi	Suwayda al Ma	186	2063	2070	2070
Al Jibayyah	Suwayda al Ma	183	2070	2070	2070
Ad Dibayshi	Sayh al Masarrat	182	2063	2063	2069

From	To	People at saturation	Donor full	Receiver starts	Receiver full
Al Akheedar	Al Makhtibyah	177	2036	2063	2069
Al Akheedar	Al Jahli	176	2036	2054	2063
Bat	Suwayda al Ma	175	2052	2070	2070
Ibri	Al Araqi	163	2056	2056	2057
Al Aynayn	Satwah	150	2060	2060	2060
Tanam	Suwayda al Ma	145	2069	2070	2070
Ibri	Miayrid	141	2056	2068	2070
Tanam	Sayh al Masarrat	141	2069	2063	2069
Sayh al Masarrat	Suwayda al Ma	125	2069	2070	2070
Al Qali	Al Jahli	103	2054	2054	2063
Hijar	Al Jibayyah	94	2053	2068	2070
Al Jahli	Al Jibayyah	92	2063	2068	2070
Al Qali	Sayh al Masarrat	83	2054	2063	2069
Al Qurayn	Al Jibayyah	78	2060	2068	2070
Hijar	Ibri	66	2053	2054	2056
Shalashil	At Tayyib	63	2061	2060	2068
Hijar	Usaybuq	60	2053	2053	2054
Hijar	Al Qali	54	2053	2036	2054
Sayh al Masarrat	Al Jibayyah	52	2069	2068	2070

A6 Saturation with and without the overflow

The year each settlement fills on its own growth alone, beside the year it fills when its neighbours' overflow is allowed to reach it. A settlement that would never fill on its own growth before 2100 is marked accordingly.

Table 35 Saturation year by settlement, own growth only and with overflow

Settlement	Capacity, people	Full on own growth	Full with overflow	People received
Ibri	73,299	2057	2056	117
Ad Dariz	16,854	2063	2062	810
Al Araqi	11,832	2057	2057	163
Al Aynayn	7,459	2066	2060	1,790
Al Wahrah	7,109	2073	2070	924
At Tayyib	34,675	not before 2100	2068	28,937
Ad Dibayshi	7,920	2083	2063	4,078
Al Jibayyah	11,306	2095	2070	6,349
Bat	2,226	2052	2052	0
Tanam	6,779	2086	2069	3,014
Sayh al Masarrat	4,984	2081	2069	1,662
Suwayda al Ma	6,369	2094	2070	3,386
Hijar	1,288	2053	2053	0
Al Jahli	3,684	2080	2063	1,743
Al Ghubayrah	3,940	not before 2100	2070	2,751
Al Qurayn	6,588	not before 2100	2060	5,856
Al Akheedar	160	2036	2036	0
Al Makhtibyah	4,292	not before 2100	2069	3,644
Al Qali	564	2078	2054	351
Shalashil	11,700	not before 2100	2061	11,503
Satwah	284	2080	2060	150
Usaybuq	140	2067	2054	60
Ash Shiab	48	2056	2056	0
Wadi al Mankas	5,020	not before 2100	2068	4,965
Miayrid	616	not before 2100	2070	586